US012398717B2

(12) United States Patent
Latham

(10) **Patent No.: US 12,398,717 B2**
(45) **Date of Patent: Aug. 26, 2025**

(54) **SINGLE PHASE FIELD ORIENTED CONTROL FOR A LINEAR COMPRESSOR**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 207 days.

(21) Appl. No.: **18/166,895**

(22) Filed: **Feb. 9, 2023**

(65) **Prior Publication Data**

US 2024/0271613 A1 Aug. 15, 2024

(51) **Int. Cl.**

F04B 49/06 (2006.01)
F04B 35/04 (2006.01)
F25B 31/02 (2006.01)

(52) **U.S. Cl.**

CPC ***F04B 49/06*** (2013.01); ***F04B 35/04*** (2013.01); ***F25B 31/023*** (2013.01)

(58) **Field of Classification Search**

CPC F04B 49/06; F04B 35/04; F04B 35/045; F04B 49/065; F25B 31/023; F25B 2600/024; F25B 2700/151; F25B 1/02; F25B 49/025

See application file for complete search history.

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Primary Examiner — Dominick L Plakkoottam

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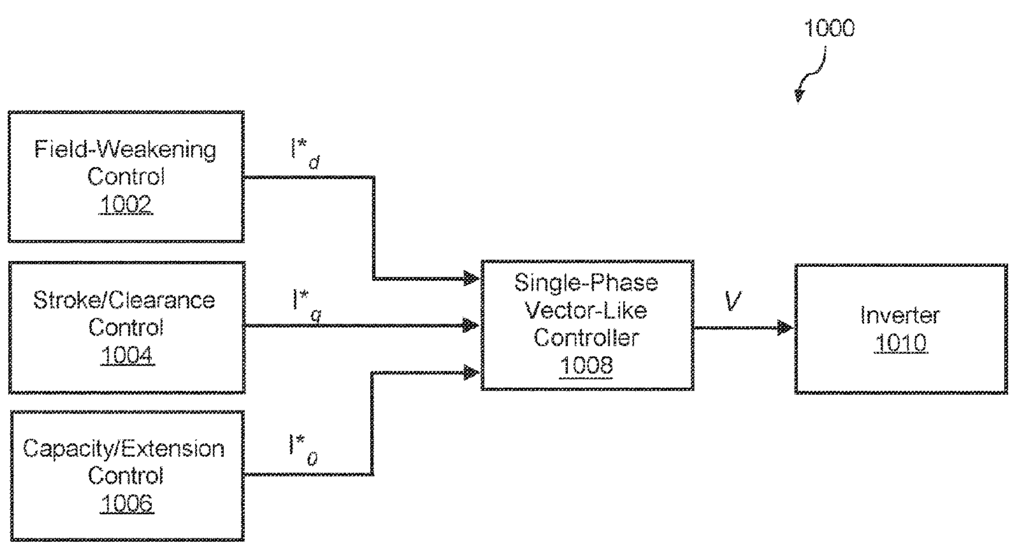
(57) **ABSTRACT**

A method for operating a linear compressor of an appliance, such as a refrigerator appliance, is provided. In one example implementation, the method can include operating a motor of the linear compressor in order to drive a rotor of the motor. The method can further include obtaining, via a controller of the linear compressor, one or more feedback measurements of one or more electrical characteristics of the motor. The method can further include controlling, based at least in part on the one or more feedback measurements, the motor of the linear compressor using a single-phase vector-like control scheme.

19 Claims, 16 Drawing Sheets



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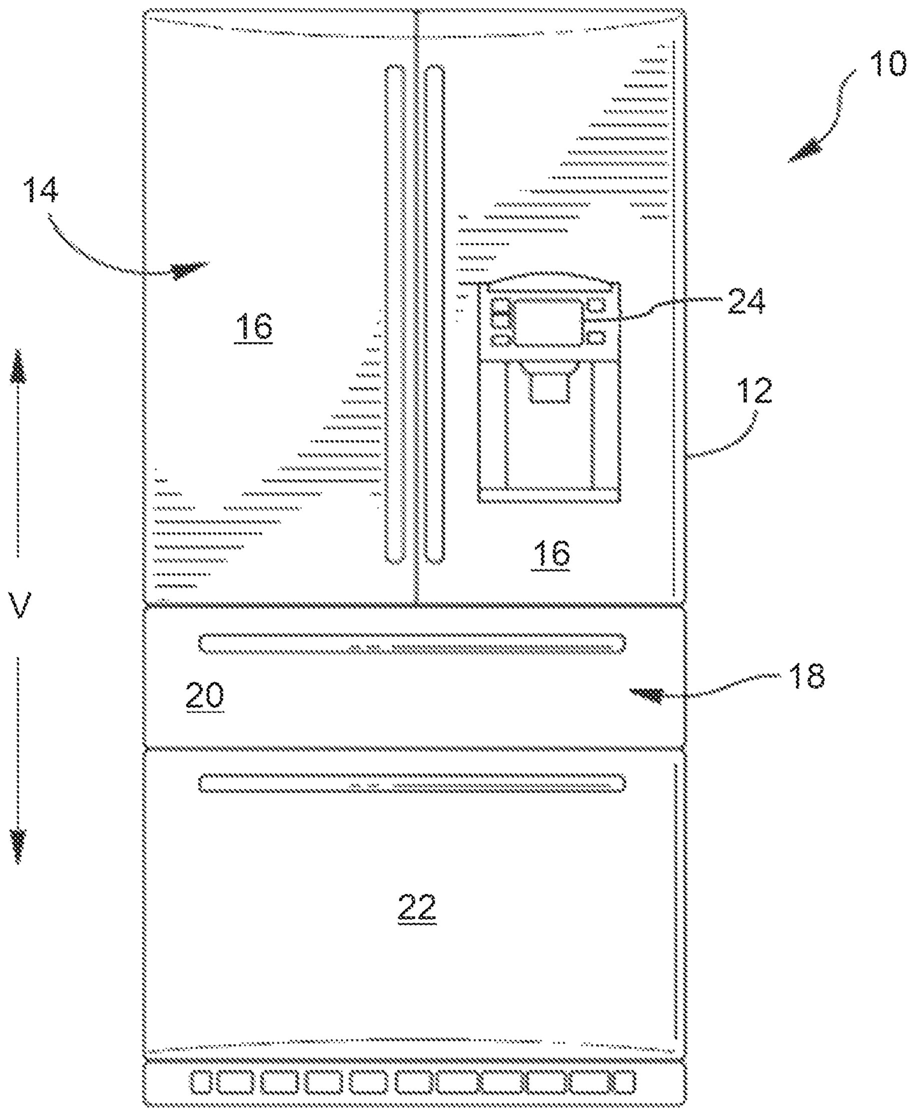


FIG. 1

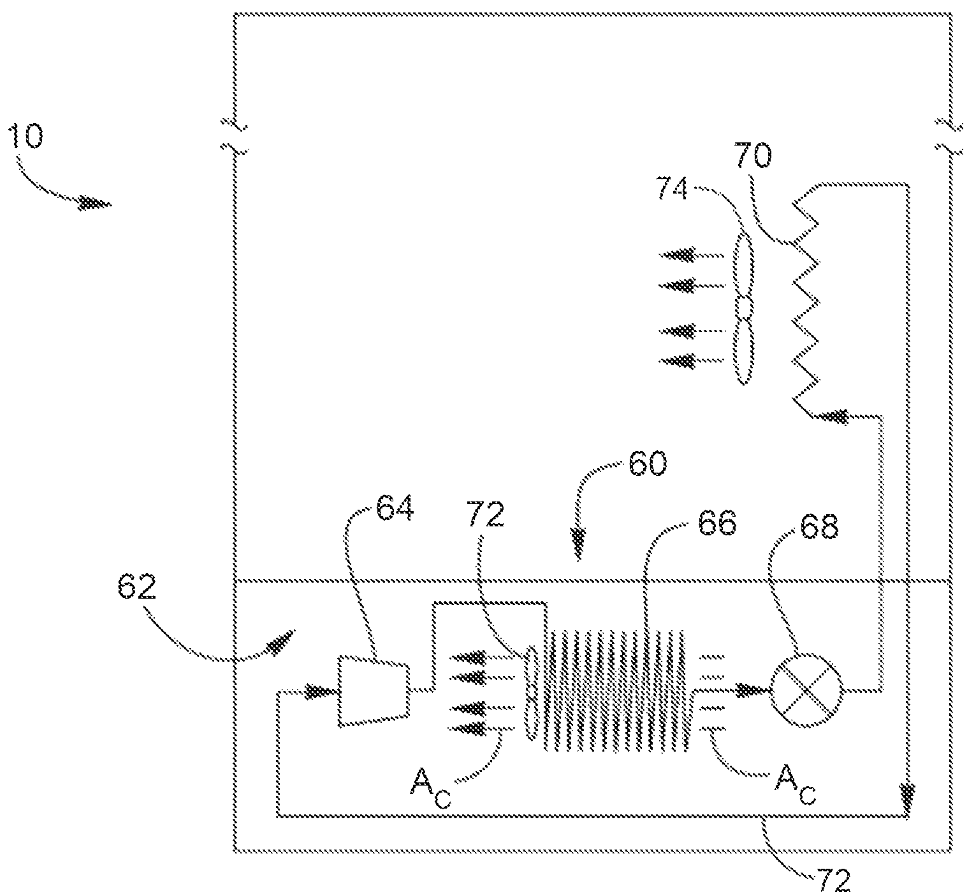
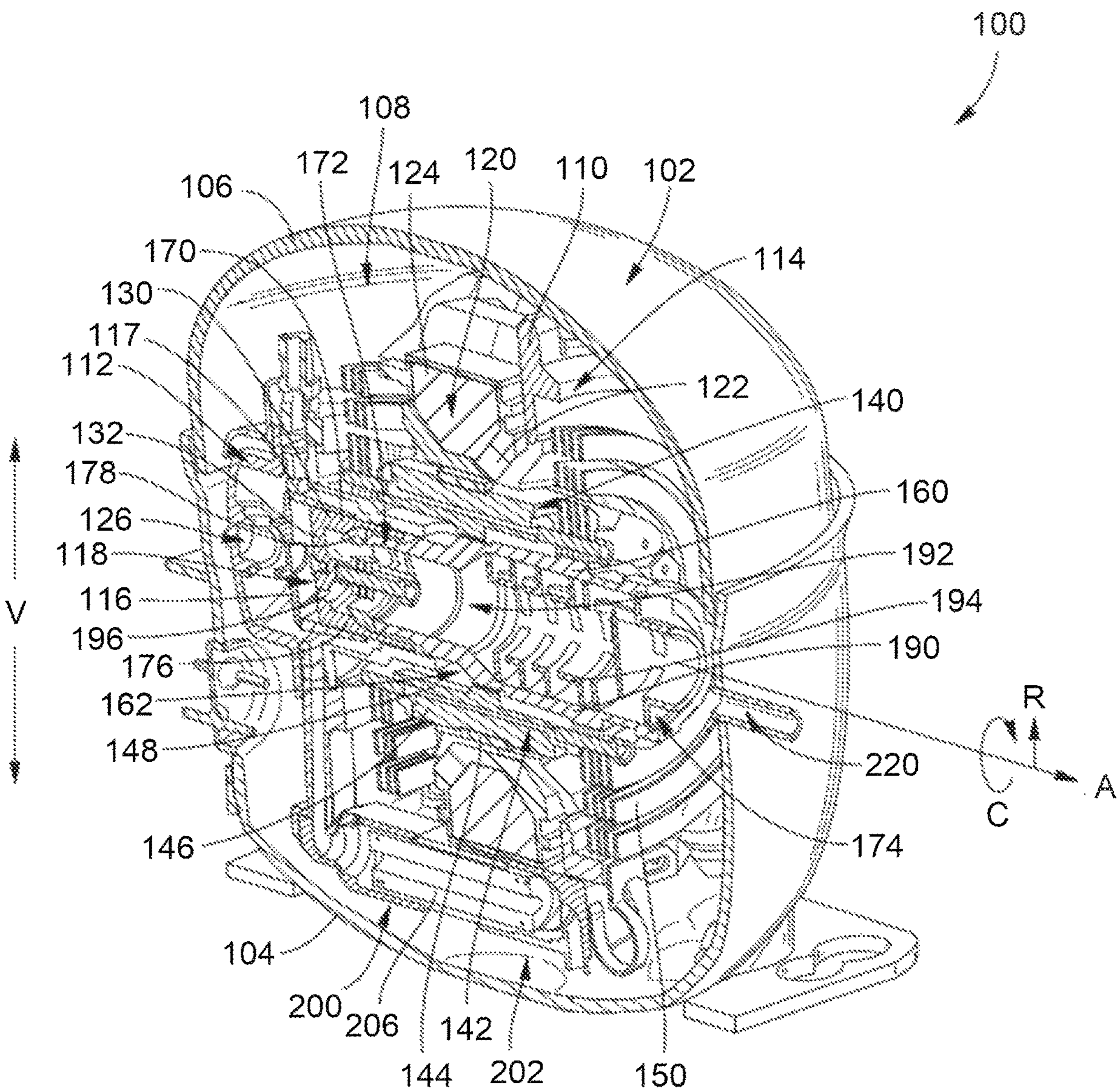


FIG. 2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192
116
194
196
V
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190
162
R
148
220
A
C
174
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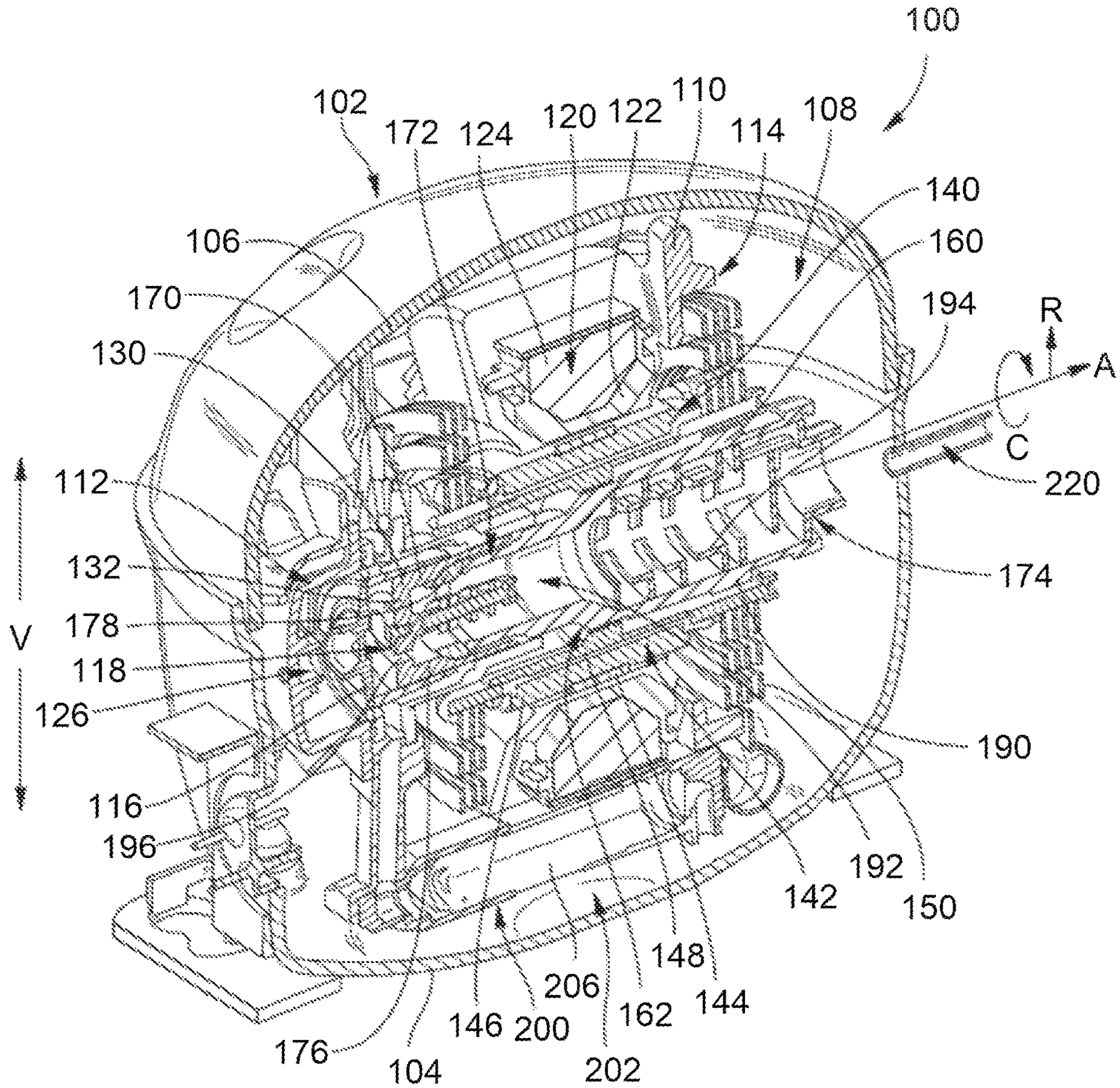
104
200
206
142
144
202
150

FIG. 3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202

FIG. 4

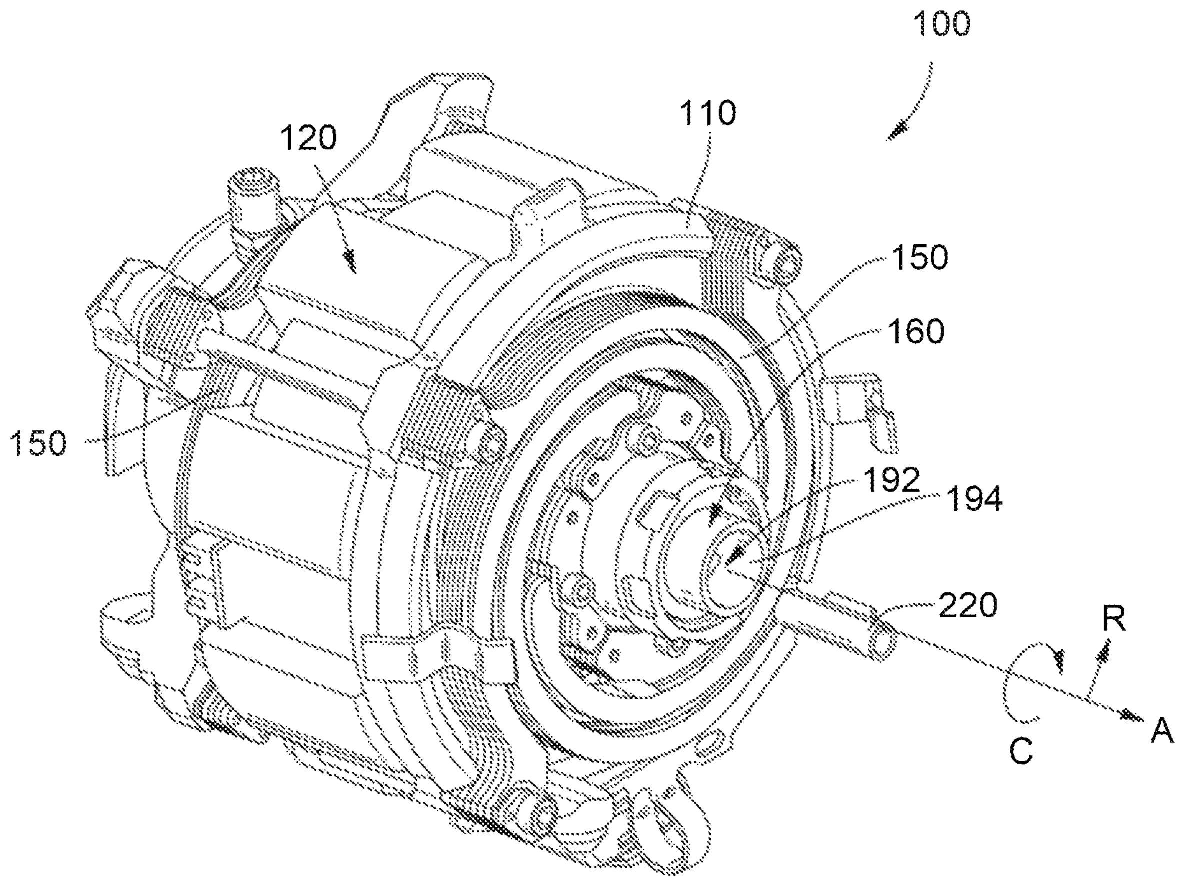
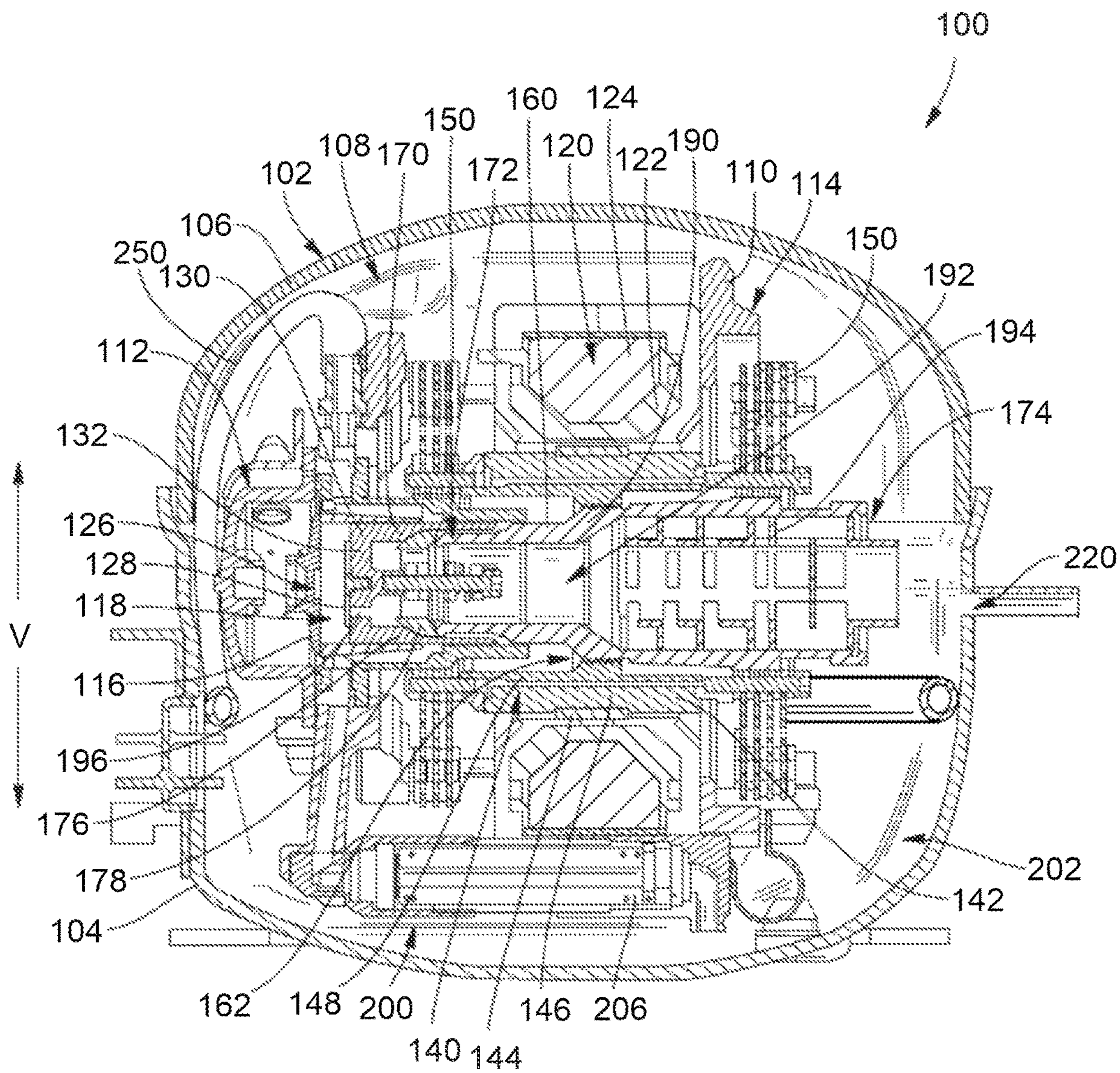


FIG. 5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174
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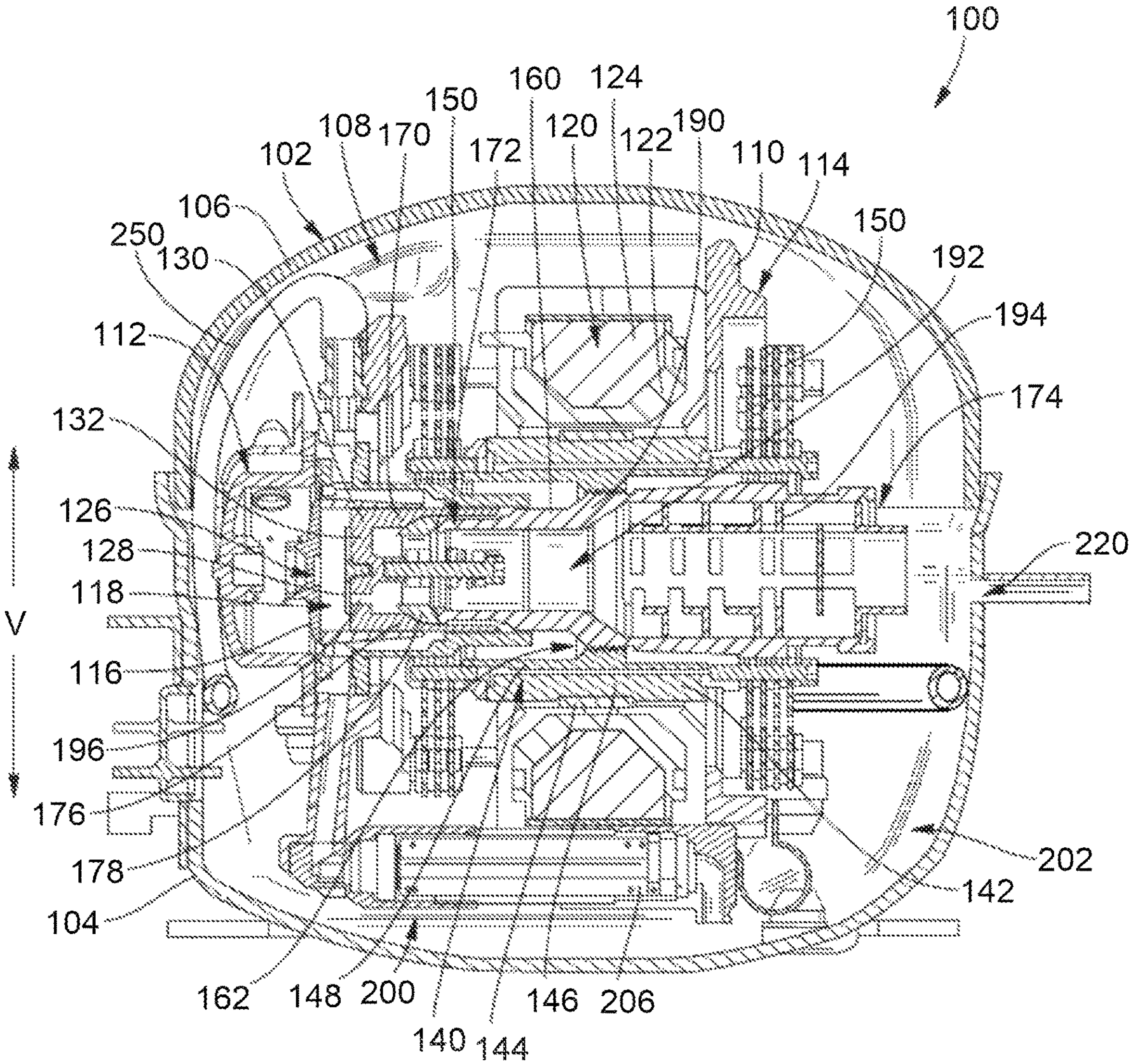
144

FIG. 6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174
126
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162
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200
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206
140
144

FIG. 7

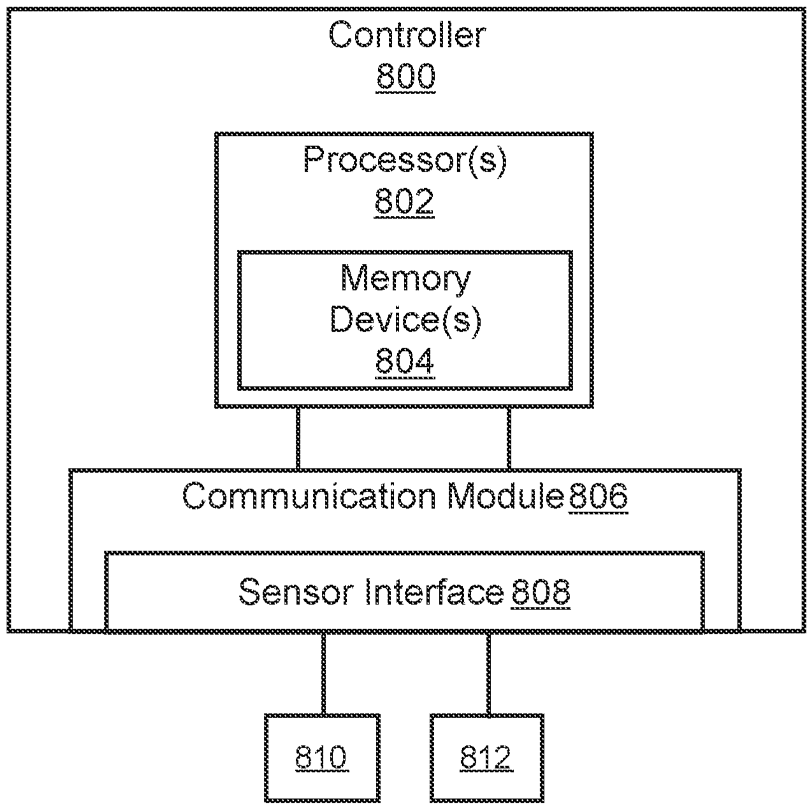
Controller
800
Processor(s)
802
Memory
Device(s)
804
Communication Module 806
Sensor Interface 808
810
812

FIG. 8

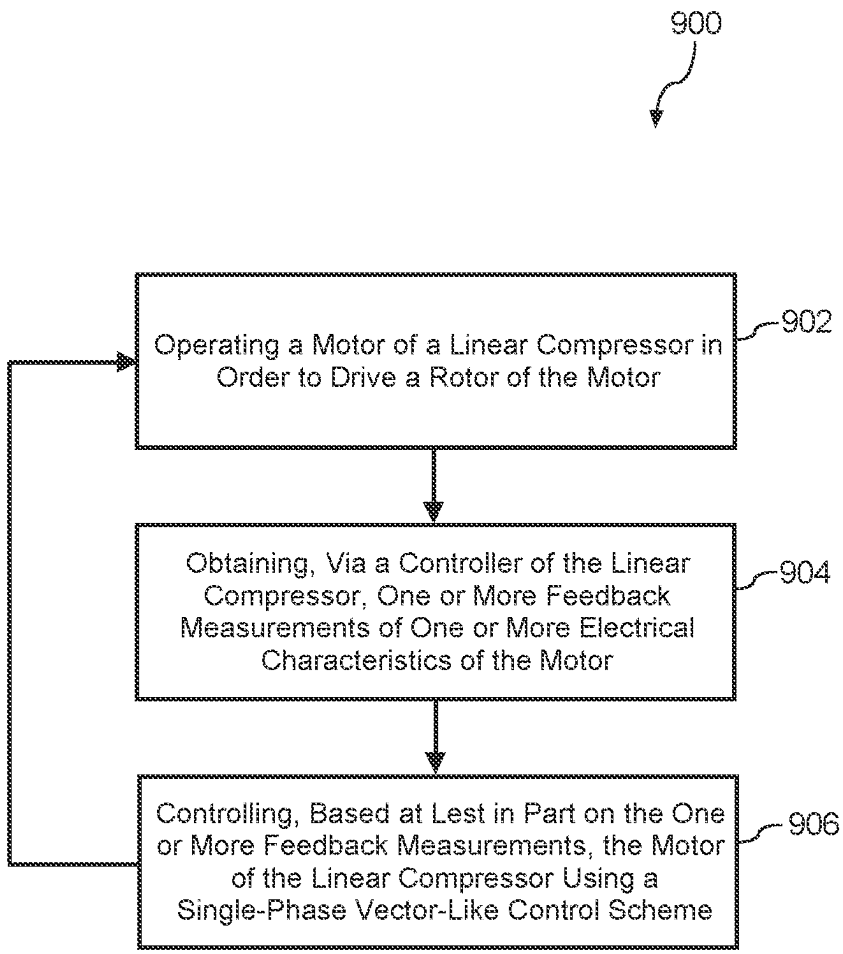
900
Operating a Motor of a Linear Compressor in Order to Drive a Rotor of the Motor
902
Obtaining, Via a Controller of the Linear Compressor, One or More Feedback Measurements of One or More Electrical Characteristics of the Motor
904
Controlling, Based at Lest in Part on the One or More Feedback Measurements, the Motor of the Linear Compressor Using a Single-Phase Vector-Like Control Scheme
906

FIG. 9

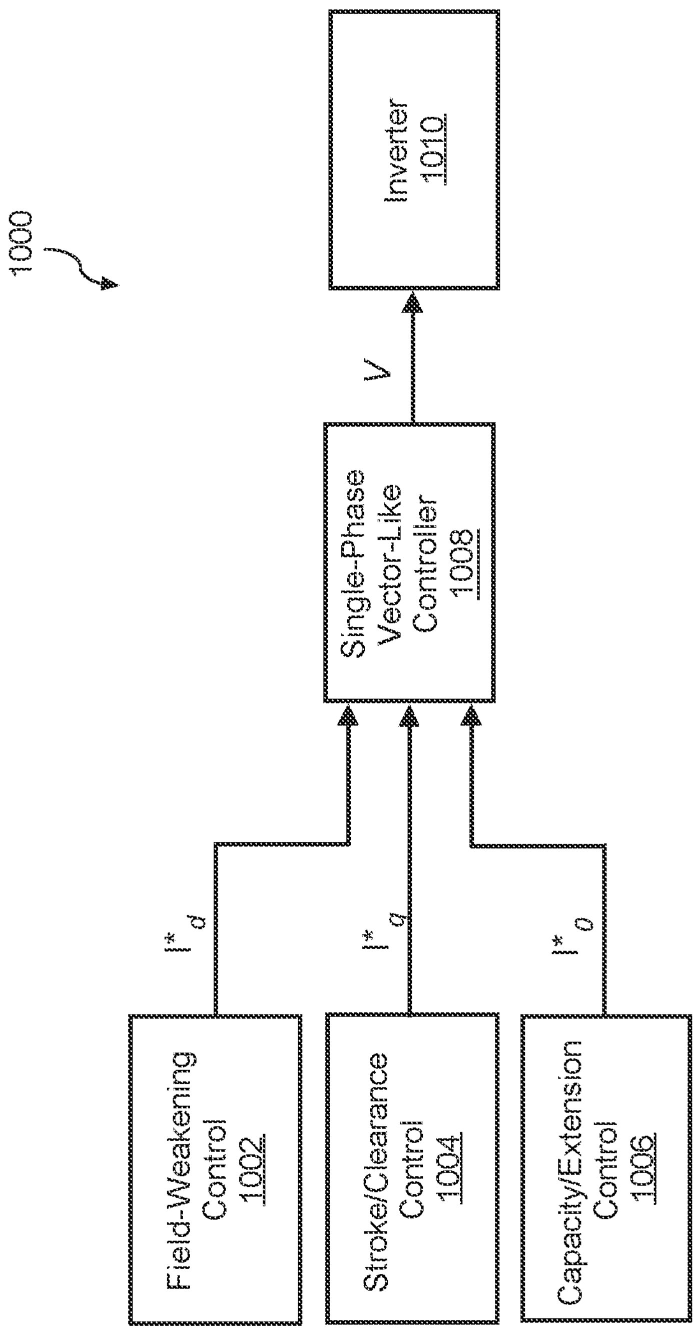
1000
Inverter
1010
V
Single-Phase
Vector-Like
Controller
1008
I*d
I*q
I*0
Field-Weakening
Control
1002
Stroke/Clearance
Control
1004
Capacity/Extension
Control
1006

FIG. 10

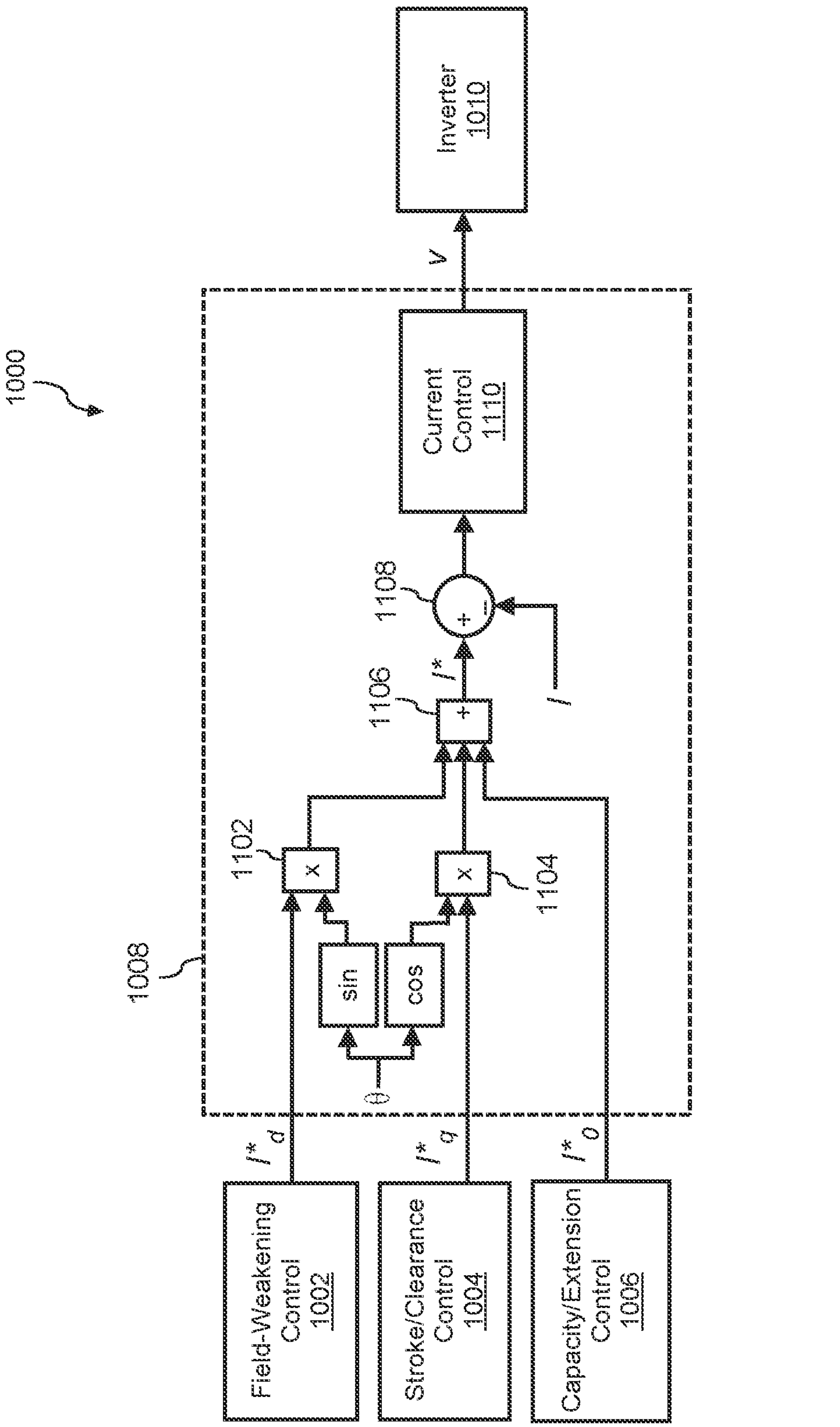
1000
Inverter
1010
V
Current Control
1110
1108
+
-
I*
1106
I
1102
x
1104
1008
sin
cos
θ
I*d
I*q
I*0
Field-Weakening Control
1002
Stroke/Clearance Control
1004
Capacity/Extension Control
1006

FIG. 11A

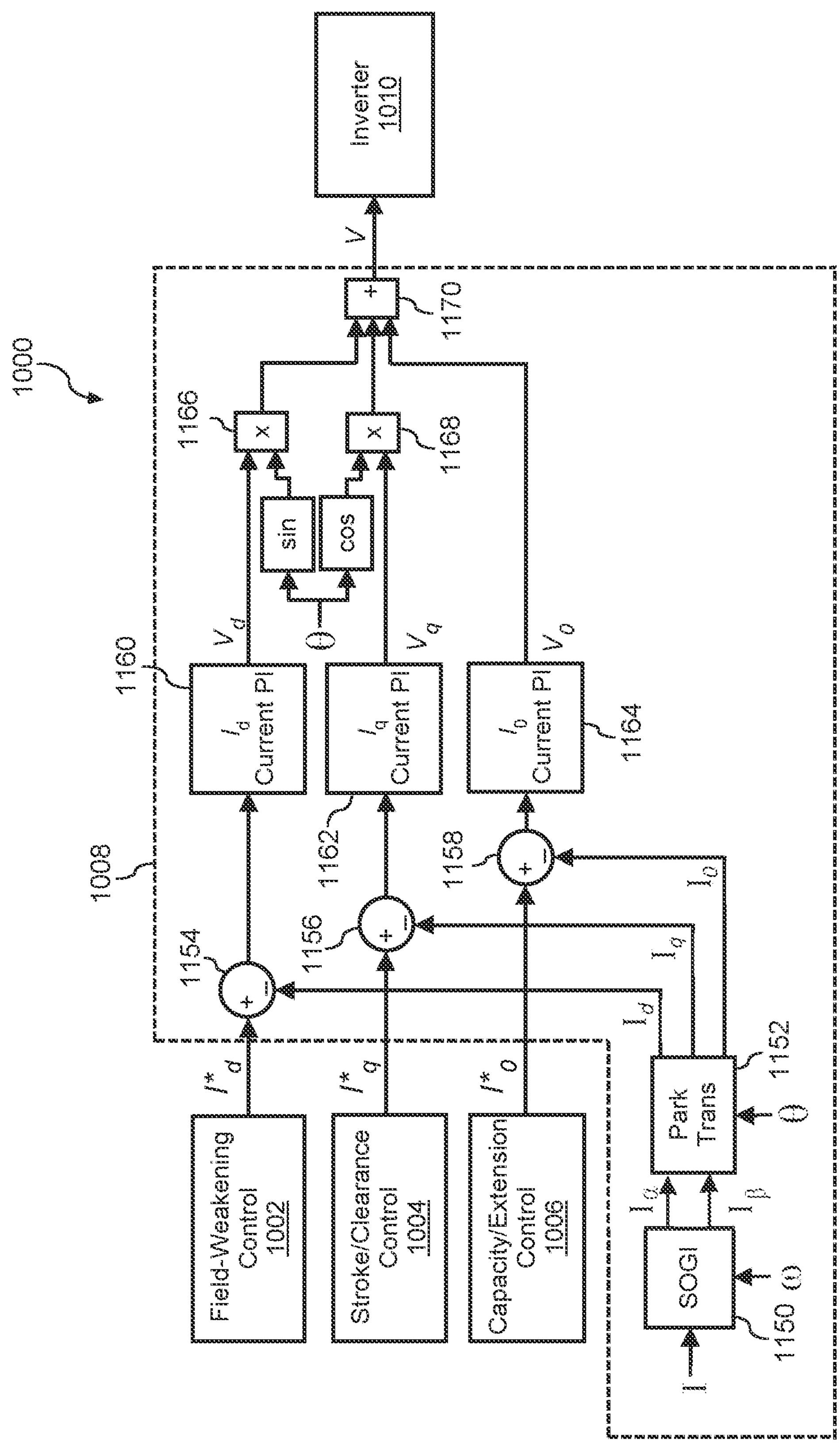
1000
Inverter
1010
V
1170
1166
1168
sin
cos
θ
V_d
V_q
V_0
1160
I_d Current PI
I_q Current PI
I_0 Current PI
1162
1164
1008
1154
1156
1158
I_d
I_q
I_0
I^*_d
I^*_q
I^*_0
Field-Weakening Control
1002
Stroke/Clearance Control
1004
Capacity/Extension Control
1006
Park Trans
1152
θ
I_α
I_β
SOGI
1150
ω
I

FIG. 11B

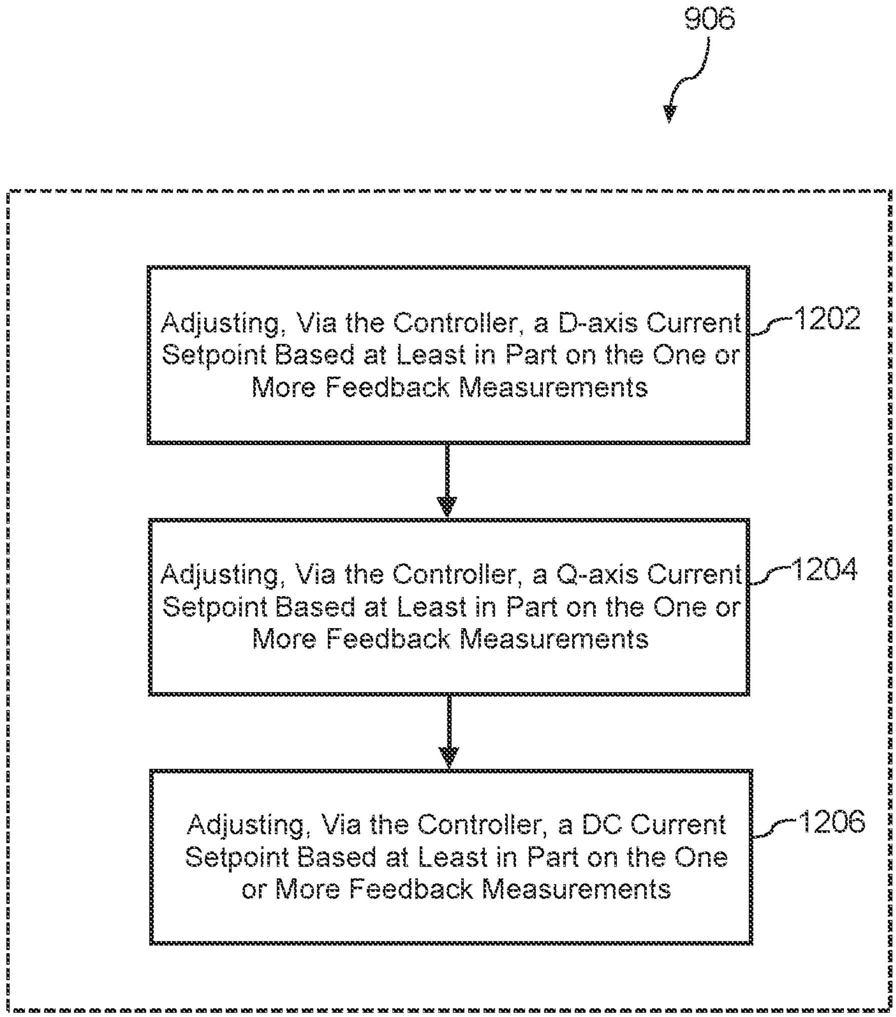
906
Adjusting, Via the Controller, a D-axis Current Setpoint Based at Least in Part on the One or More Feedback Measurements
1202
Adjusting, Via the Controller, a Q-axis Current Setpoint Based at Least in Part on the One or More Feedback Measurements
1204
Adjusting, Via the Controller, a DC Current Setpoint Based at Least in Part on the One or More Feedback Measurements
1206

FIG. 12

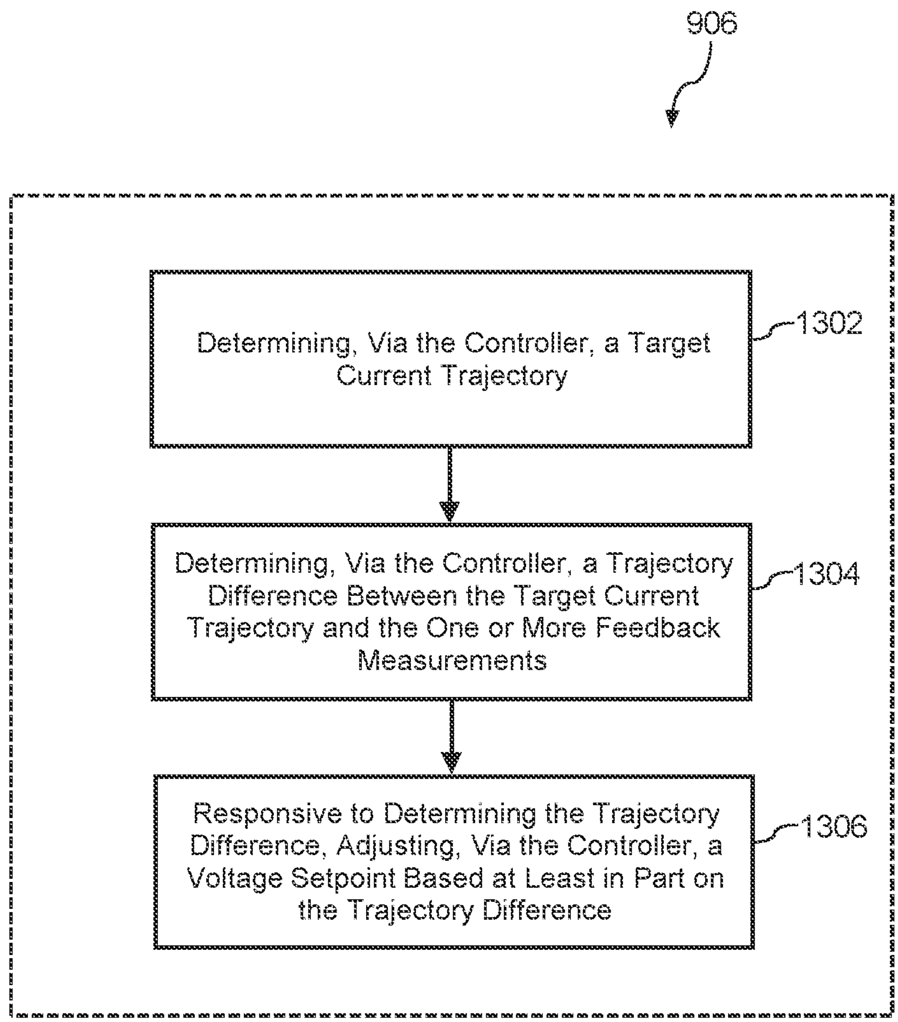
906
Determining, Via the Controller, a Target Current Trajectory
1302
Determining, Via the Controller, a Trajectory Difference Between the Target Current Trajectory and the One or More Feedback Measurements
1304
Responsive to Determining the Trajectory Difference, Adjusting, Via the Controller, a Voltage Setpoint Based at Least in Part on the Trajectory Difference
1306

FIG. 13

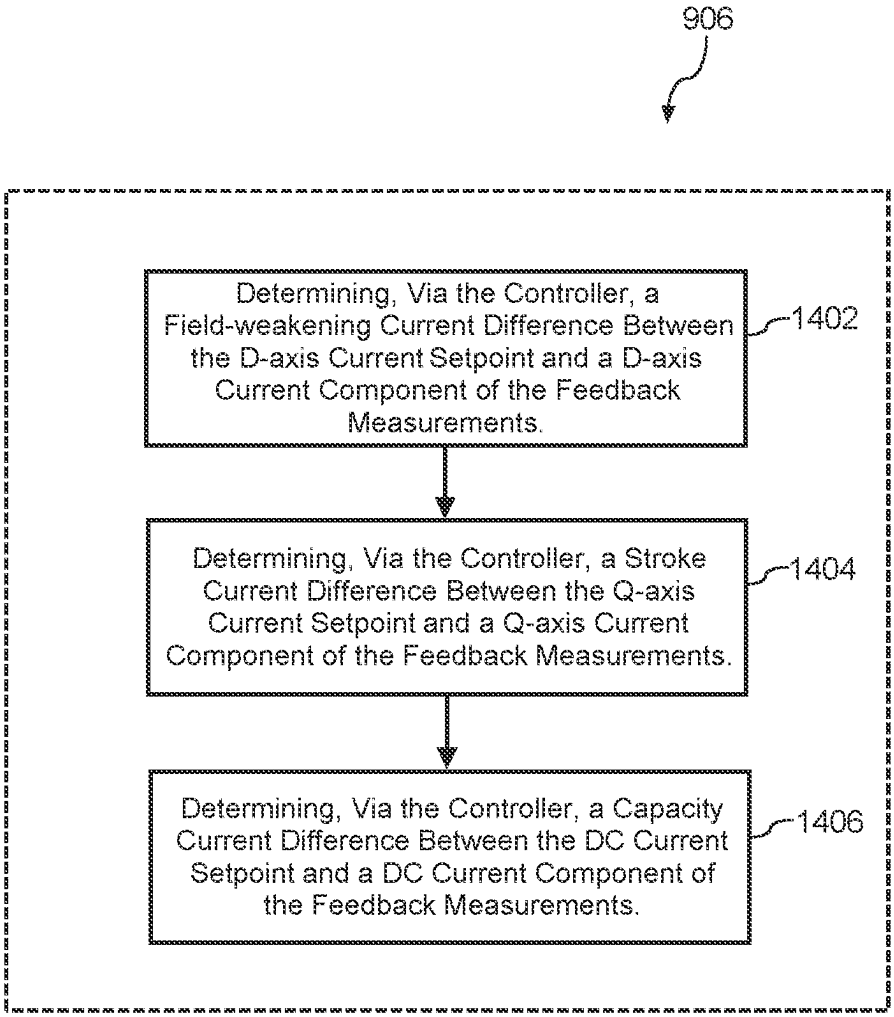
906
Determining, Via the Controller, a Field-weakening Current Difference Between the D-axis Current Setpoint and a D-axis Current Component of the Feedback Measurements.
1402
Determining, Via the Controller, a Stroke Current Difference Between the Q-axis Current Setpoint and a Q-axis Current Component of the Feedback Measurements.
1404
Determining, Via the Controller, a Capacity Current Difference Between the DC Current Setpoint and a DC Current Component of the Feedback Measurements.
1406

FIG. 14

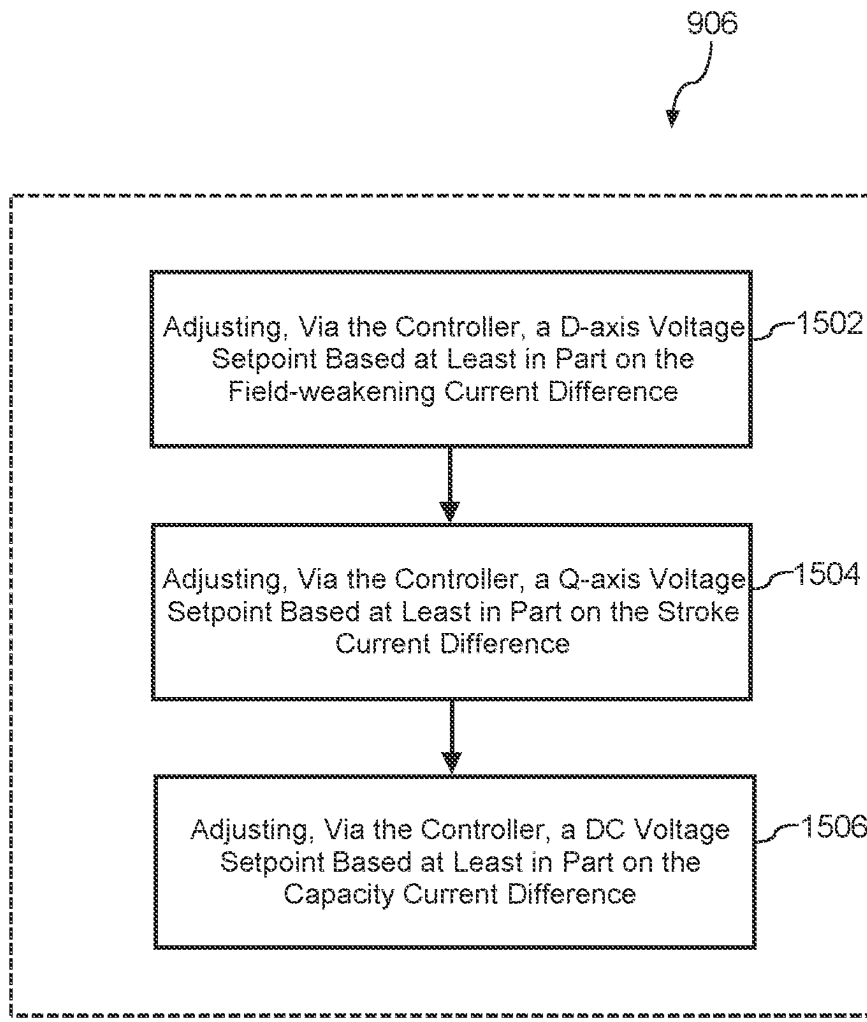
906
Adjusting, Via the Controller, a D-axis Voltage Setpoint Based at Least in Part on the Field-weakening Current Difference
1502
Adjusting, Via the Controller, a Q-axis Voltage Setpoint Based at Least in Part on the Stroke Current Difference
1504
Adjusting, Via the Controller, a DC Voltage Setpoint Based at Least in Part on the Capacity Current Difference
1506

FIG. 15

SINGLE PHASE FIELD ORIENTED CONTROL FOR A LINEAR COMPRESSOR

FIELD

Example aspects of the present disclosure relate to linear compressors, such as linear compressors for refrigerators and other appliances.

BACKGROUND

Generally, refrigerator appliances include a cabinet that defines one or more chilled chambers, such as a fresh food chamber for receipt of food items for storage and/or a freezer chamber for receipt of food items for freezing and storage. Certain refrigerator appliances may also include sealed systems for cooling such chilled chambers thereof. The sealed systems generally include a compressor that generates compressed refrigerant during operation thereof. The compressed refrigerant flows to an evaporator where heat exchanges between the chilled chambers and the refrigerant cools the chilled chambers and food items located therein.

SUMMARY

Aspects and advantages of embodiments of the present disclosure will be set forth in part in the following description, or can be learned from the description, or can be learned through practice of the embodiments.

One example aspect of the present disclosure is directed to an appliance (e.g., a refrigerator appliance) having a cabinet defining an internal chamber, a door rotatably mounted to the cabinet to provide selective access to the internal chamber, a linear compressor having a piston, a motor operably coupled to the piston, and an inverter. The piston of the linear compressor can be movable in a negative axial direction toward a compressor chamber and in a positive axial direction away from the compressor chamber. The inverter can be configured to supply a variable frequency waveform to the motor. The appliance can also include a controller operably coupled to the motor. The controller can be configured to operate the motor of the appliance in order to drive a rotor of the motor. The controller can be further configured to obtain one or more feedback measurements of one or more electrical characteristics of the motor. The controller can be further configured to control the motor using a single-phase vector-like control scheme based at least in part on the one or more feedback measurements.

These and other features, aspects and advantages of various embodiments will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the present disclosure and, together with the description, serve to explain the related principles.

BRIEF DESCRIPTION OF THE DRAWINGS

Detailed discussion of embodiments directed to one of ordinary skill in the art are set forth in the specification, which makes reference to the appended figures, in which:

FIG. **1** depicts a front elevation view of a refrigerator appliance according to example embodiments of the present disclosure;

FIG. **2** depicts a schematic view of certain components of the example refrigerator appliance of FIG. **1** according to example embodiments of the present disclosure;

FIG. **3** depicts a perspective, section view of a linear compressor according to an exemplary embodiment of the present subject matter;

FIG. **4** depicts another perspective, section view of the exemplary linear compressor of FIG. **3** according to an embodiment of the present subject matter;

FIG. **5** depicts a perspective view of a linear compressor with a compressor housing removed for clarity according to an example embodiment of the present subject matter;

FIG. **6** depicts a section view of the exemplary linear compressor of FIG. **3** with a piston in an extended position according to an embodiment of the present subject matter;

FIG. **7** depicts a section view of the exemplary linear compressor of FIG. **3** with the piston in a retracted position according to an embodiment of the present subject matter;

FIG. **8** depicts a block diagram of one embodiment of a controller of a refrigerator appliance according to example embodiments of the present disclosure;

FIG. **9** depicts an example method for operating a linear compressor according to example embodiments of the present disclosure;

FIG. **10** depicts an example control scheme operable to control a linear compressor according to example embodiments of the present disclosure.

FIG. **11**A depicts an example single-phase vector-like control scheme implemented by the controller of FIG. **10** according to example embodiments of the present disclosure;

FIG. **11**B depicts an example single-phase vector-like control scheme implemented by the controller of FIG. **10** according to example embodiments of the present disclosure;

FIG. **12** depicts an example method for operating a linear compressor according to example embodiments of the present disclosure;

FIG. **13** depicts an example method for operating a linear compressor according to example embodiments of the present disclosure;

FIG. **14** depicts an example method for operating a linear compressor according to example embodiments of the present disclosure; and

FIG. **15** depicts an example method for operating a linear compressor according to example embodiments of the present disclosure.

Repeat use of reference characters in the present specification and drawings is intended to represent the same and/or analogous features or elements of the present invention.

DETAILED DESCRIPTION

Reference now will be made in detail to embodiments, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the embodiments, not limitation of the present disclosure. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made to the embodiments without departing from the scope or spirit of the present disclosure. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that aspects of the present disclosure cover such modifications and variations.

Example aspects of the present disclosure relate generally to linear compressors such as, e.g., linear compressors for refrigerators and other appliances.

Recently, certain refrigerator appliances have included linear compressors for compressing the refrigerant. Linear compressors generally include a piston within a housing and a driving coil that generates a force for moving the piston forward and backward within the housing. During motion of the piston within the housing, the piston compresses the refrigerant. Furthermore, linear compressors are generally operated a single-phase motor driven by a single-phase variable-frequency drive. The variable-frequency drive is a type of motor drive that is used to control the motor speed and force by varying motor voltage input frequency and amplitude.

In order to optimally drive the linear compressor, the phase between motor current and back electromotive force (back-EMF) of the motor must be manipulated. This same principle can apply to three-phase motors (e.g., brushless direct current (BLDC) motors, permanent magnet synchronous motors (PMSM)). In the case of three-phase motors, the phase between motor current and back-EMF can be controlled by a controller implementing a field-oriented control (FOC) control scheme. FOC control schemes define two components of a target current of the motor-a direct current (i.e., d-axis current) component and a quadrature current (e.g., q-axis current) component. Furthermore, FOC control schemes may provide high efficiency for and high-fidelity speed and/or position control. However, FOC control schemes are implemented in three-phase control systems. Accordingly, a linear compressor implementing a single-phase control scheme similar to a three-phase FOC control scheme is desired.

According to example aspects of the present disclosure, an appliance (e.g., a refrigerator appliance) can include a single-phase linear compressor driven by a single-phase linear motor. The linear compressor can implement an FOC-like control scheme, such as a single-phase vector-like control scheme, to control the motor based at least in part on one or more feedback measurements of one or more electrical characteristics (e.g., current, voltage, flux) of the motor.

The systems and methods according to example embodiments of the present disclosure provide a number of technical effects and benefits. For instance, example aspects of the present disclosure provide a single-phase vector-like control scheme similar to a three-phase field-oriented control scheme. Furthermore, by implementing the single-phase vector-like control scheme, example aspects of the present disclosure provide stable and robust control of the linear compressor. Additionally, example aspects of the present disclosure allow for decoupling of stroke and/or field-weakening control, thereby providing a higher degree of control than that provided by conventional control schemes for single-phase linear compressors. In this way, example aspects of the present disclosure provide more flexibility in the control algorithms that can be implemented to control the linear compressor.

As used herein, the terms "first," "second," and "third" may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components. The terms "includes" and "including" are intended to be inclusive in a manner similar to the term "comprising." Similarly, the term "or" is generally intended to be inclusive (e.g., "A or B" is intended to mean "A or B or both"). The term "at least one of" in the context of, e.g., "at least one of A, B, and C" refers to only A, only B, only C, or any combination of A, B, and C. In addition, here and throughout the specification and claims, range limitations may be combined and/or interchanged. Such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other. The singular forms "a," "an," and "the" include plural references unless the context clearly dictates otherwise.

Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as "generally," "about," "approximately," and "substantially," are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a 10 percent margin, i.e., including values within ten percent greater or less than the stated value. In this regard, for example, when used in the context of an angle or direction, such terms include within ten degrees greater or less than the stated angle or direction, e.g., "generally vertical" includes forming an angle of up to ten degrees in any direction, e.g., clockwise or counterclockwise, with the vertical direction V.

The word "exemplary" is used herein to mean "serving as an example, instance, or illustration." In addition, references to "an embodiment" or "one embodiment" does not necessarily refer to the same embodiment, although it may. Any implementation described herein as "exemplary" or "an embodiment" is not necessarily to be construed as preferred or advantageous over other implementations. Moreover, each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

FIG. **1** depicts a refrigerator appliance **10** that incorporates a sealed refrigeration system **60** (FIG. **2**). It should be appreciated that the term "refrigerator appliance" is used in a generic sense herein to encompass any manner of refrigeration appliance, such as a freezer, refrigerator/freezer combination, and any style or model of conventional refrigerator. In addition, it should be understood that the present subject matter is not limited to use in refrigerator appliances. Thus, the present subject matter may be used for any other suitable purpose, such as vapor compression within air conditioning units or air compression within air compressors.

In the illustrated example embodiment shown in FIG. **1**, the refrigerator appliance **10** is depicted as an upright refrigerator having a cabinet **12** that defines a number of internal chilled storage compartments. In particular, refrigerator appliance **10** includes upper fresh-food compartments **14** having doors **16** and lower freezer compartment **18** having upper drawer **20** and lower drawer **22**. The drawers **20** and **22** are "pull-out" drawers in that they can be manually moved into and out of the freezer compartment **18** on suitable slide mechanisms. Due to the doors **16** being rotatably mounted to the cabinet **12**, the doors **16** provide selective access to the internal chamber. As used herein, "internal chamber" refers to the internal chilled storage compartments defined by the cabinet **12**.

FIG. **2** is a schematic view of certain components of refrigerator appliance **10**, including a sealed refrigeration system **60** of refrigerator appliance **10**. A machinery compartment **62** contains components for executing a known vapor compression cycle for cooling air. The components include a compressor **64**, a condenser **66**, an expansion device **68**, and an evaporator **70** connected in series by fluid conduit **72** that is charged with a refrigerant. As will be understood by those skilled in the art, refrigeration system **60** may include additional components, e.g., at least one additional evaporator, compressor, expansion device, and/or condenser. As an example, refrigeration system **60** may include two evaporators.

Within refrigeration system **60**, refrigerant flows into compressor **64**, which operates to increase the pressure of the refrigerant. This compression of the refrigerant raises its temperature, which is lowered by passing the refrigerant through condenser **66**. Within condenser **66**, heat exchange with ambient air takes place so as to cool the refrigerant. A fan **74** is used to pull air across condenser **66**, as illustrated by arrows A_C, so as to provide forced convection for a more rapid and efficient heat exchange between the refrigerant within condenser **66** and the ambient air. Thus, as will be understood by those skilled in the art, increasing air flow across condenser **66** can, e.g., increase the efficiency of condenser **66** by improving cooling of the refrigerant contained therein.

An expansion device **68** (e.g., a valve, capillary tube, or other restriction device) receives refrigerant from condenser **66**. From expansion device **68**, the refrigerant enters evaporator **70**. Upon exiting expansion device **68** and entering evaporator **70**, the refrigerant drops in pressure. Due to the pressure drop and/or phase change of the refrigerant, evaporator **70** is cool relative to compartments **14** and **18** of refrigerator appliance **10**. As such, cooled air is produced and refrigerates compartments **14** and **18** of refrigerator appliance **10**. Thus, evaporator **70** is a type of heat exchanger which transfers heat from air passing over evaporator **70** to refrigerant flowing through evaporator **70**.

Collectively, the vapor compression cycle components in a refrigeration circuit, associated fans, and associated compartments are sometimes referred to as a sealed refrigeration system operable to force cold air through compartments **14**, **18** (FIG. **1**). The refrigeration system **60** depicted in FIG. **2** is provided by way of example only. Thus, it is within the scope of the present subject matter for other configurations of the refrigeration system to be used as well.

As described above, sealed refrigeration system **60** performs a vapor compression cycle to refrigerate compartments **14**, **18** of refrigerator appliance **10**. However, as is understood in the art, refrigeration system **60** is a sealed system that may be alternately operated as a refrigeration assembly (and thus perform a refrigeration cycle as described above) or a heat pump (and thus perform a heat pump cycle). Thus, for example, aspects of the present subject matter may similarly be used in a sealed system for an air conditioner unit, e.g., to perform by a refrigeration or cooling cycle and a heat pump or heating cycle. In this regard, when a sealed system is operating in a cooling mode and thus performs a refrigeration cycle, an indoor heat exchanger acts as an evaporator and an outdoor heat exchanger acts as a condenser. Alternatively, when the sealed system is operating in a heating mode and thus performs a heat pump cycle, the indoor heat exchanger acts as a condenser and the outdoor heat exchanger acts as an evaporator.

Referring now generally to FIGS. **3** through **7**, a linear compressor **100** is described according to exemplary embodiments of the present subject matter. Specifically, FIGS. **3** and **4** provide perspective, section views of the linear compressor **100**, FIG. **5** provides a perspective view of the linear compressor **100** with a compressor shell or housing **102** removed for clarity, and FIGS. **6** and **7** provide section views of the linear compressor when a piston thereof is in extended and retracted positions, respectively. It should be appreciated that the linear compressor **100** is used herein only as an exemplary embodiment to facilitate the description of aspects of the present subject matter. Modifications and variations may be made to the linear compressor **100** while remaining within the scope of the present subject matter.

As illustrated for example in FIGS. **3** and **4**, the housing **102** may include a lower portion or lower housing **104** and an upper portion or upper housing **106** which are joined together to form a substantially enclosed cavity **108** for housing various components of linear compressor **100**. Specifically, for example, cavity **108** may be a hermetic or air-tight shell that can house working components of linear compressor **100** and may hinder or prevent refrigerant from leaking or escaping from refrigeration system **60**. In addition, linear compressor **100** generally defines an axial direction A, a radial direction R, and a circumferential direction C. It should be appreciated that linear compressor **100** is described and illustrated herein only to describe aspects of the present subject matter. Variations and modifications to linear compressor **100** may be made while remaining within the scope of the present subject matter.

Referring particularly to FIGS. **3** through **7**, various parts and working components of the linear compressor **100** will be described according to an exemplary embodiment. As shown, the linear compressor **100** includes a casing **110** that extends between a first end portion **112** and a second end portion **114**, e.g., along the axial direction A. The casing **110** includes a cylinder **117** that defines a compressor chamber **118**. The cylinder **117** is positioned at or adjacent first end portion **112** of casing **110**. The chamber **118** extends longitudinally along the axial direction A. As discussed in greater detail below, the linear compressor **100** is operable to increase a pressure of fluid within chamber **118** of linear compressor **100**. Further, the linear compressor **100** may be used to compress any suitable fluid, such as refrigerant or air. In particular, the linear compressor **100** may be used in a refrigerator appliance, such as refrigerator appliance **10** (FIG. **1**) in which the linear compressor **100** may be used as compressor **64** (FIG. **2**).

Moreover, as shown, the linear compressor **100** includes a stator **120** of a motor that is mounted or secured to casing **110**. For example, stator **120** generally includes an outer back iron **122** and a driving coil **124** that extend about the circumferential direction C within casing **110**. The linear compressor **100** also includes one or more valves that permit refrigerant to enter and exit chamber **118** during operation of linear compressor **100**. For example, a discharge muffler **126** is positioned at an end of chamber **118** for regulating the flow of refrigerant out of chamber **118**, while a suction valve **128** (shown only in FIGS. **6-7** for clarity) regulates flow of refrigerant into chamber **118**.

A piston **130** with a piston head **132** is slidably received within chamber **118** of cylinder **117**. The piston **130** can be operably coupled to the motor. In particular, piston **130** is movable along the axial direction A. For instance, the piston **130** can be movable in a negative axial direction A toward the chamber **118**. The piston **130** can be movable in a positive axial direction A away from the chamber **118**. During sliding of piston head **132** within chamber **118**, piston head **132** compresses refrigerant within chamber **118**. As an example, from a top dead center position (see, e.g., FIG. **6**), piston head **132** can slide within chamber **118** towards a bottom dead center position (see, e.g., FIG. **7**) along the axial direction A, i.e., an expansion stroke of piston head **132**. When piston head **132** reaches the bottom dead center position, piston head **132** changes directions and slides in chamber **118** back towards the top dead center position, i.e., a compression stroke of piston head **132**. It should be understood that the linear compressor **100** may include an additional piston head and/or additional chambers at an opposite end of linear compressor **100**. Thus, linear compressor **100** may have multiple piston heads in alternative exemplary embodiments.

As illustrated, the linear compressor **100** also includes a mover **140** which is generally driven by stator **120** for compressing refrigerant. Specifically, for example, mover **140** may include an inner back iron **142** positioned in stator **120** of the motor. In particular, outer back iron **122** and/or driving coil **124** may extend about inner back iron **142**, e.g., along the circumferential direction C. Inner back iron **142** also has an outer surface that faces towards outer back iron **122** and/or driving coil **124**. At least one driving magnet **144** is mounted to inner back iron **142**, e.g., at the outer surface of inner back iron **142**.

Driving magnet **144** may face and/or be exposed to driving coil **124**. In particular, driving magnet **144** may be spaced apart from driving coil **124**, e.g., along the radial direction R by an air gap. Thus, the air gap may be defined between opposing surfaces of driving magnet **144** and driving coil **124**. Driving magnet **144** may also be mounted or fixed to inner back iron **142** such that an outer surface of driving magnet **144** is substantially flush with the outer surface of inner back iron **142**. Thus, driving magnet **144** may be inset within inner back iron **142**. In such a manner, the magnetic field from driving coil **124** may have to pass through only a single air gap between outer back iron **122** and inner back iron **142** during operation of the linear compressor **100**, and the linear compressor **100** may be more efficient relative to linear compressors with air gaps on both sides of a driving magnet.

As may be seen in FIG. **3**, the driving coil **124** extends about inner back iron **142**, e.g., along the circumferential direction C. In alternative example embodiments, inner back iron **142** may extend around driving coil **124** along the circumferential direction C. The driving coil **124** is operable to move the inner back iron **142** along the axial direction A during operation of driving coil **124**. As an example, a current may be induced within driving coil **124** by a current source (not shown) to generate a magnetic field that engages driving magnet **144** and urges piston **130** to move along the axial direction A in order to compress refrigerant within chamber **118** as described above and will be understood by those skilled in the art. In particular, the magnetic field of driving coil **124** may engage driving magnet **144** in order to move inner back iron **142** and piston head **132** along the axial direction A during operation of driving coil **124**. Thus, the driving coil **124** may slide the piston **130** between the top dead center position and the bottom dead center position, e.g., by moving inner back iron **142** along the axial direction A, during operation of driving coil **124**.

Referring particularly to FIG. **8**, operation of the refrigerator appliance **10** may generally be controlled by a processing device or controller **800**. The controller **800** may, for example, be operatively coupled to the control panel **24** for user manipulation to select features and operations of the refrigerator appliance **10**, such as temperature set points. Thus, the controller **800** can operate various components of the refrigerator appliance **10** to execute selected system cycles, processes, and/or features. In exemplary embodiments, the controller **800** is in operative communication (e.g., electrical or wireless communication) with each of the chambers or compartments therein, for example, to regulate temperature as described herein.

More specifically, as shown in FIG. **8**, a block diagram of one embodiment of suitable components that may be included within the controller **800** in accordance with example aspects of the present disclosure is illustrated. As shown, the controller **800** may include one or more processor(s) **802**, computer, or other suitable processing unit and associated memory device(s) **804** that may include suitable computer-readable instructions that, when implemented, configure the controller to perform various different functions, such as receiving, transmitting and/or executing signals (e.g., performing the methods, steps, calculations and the like disclosed herein).

As used herein, the term "processor" refers not only to integrated circuits referred to in the art as being included in a computer, but also refers to a controller, a microcontroller, a microcomputer, a programmable logic controller (PLC), an application specific integrated circuit, and other programmable circuits. Additionally, the memory device(s) **804** may generally include memory element(s) including, but not limited to, computer readable medium (e.g., random access memory (RAM)), computer readable non-volatile medium (e.g., a flash memory), a floppy disk, a compact disc-read only memory (CD-ROM), a magneto-optical disk (MOD), a digital versatile disc (DVD) and/or other suitable memory elements. The memory can be a separate component from the processor or can be included onboard within the processor.

Such memory device(s) **804** may generally be configured to store suitable computer-readable instructions that, when implemented by the processor(s) **802**, configure the controller to perform various functions as described herein. In particular, the processor(s) **802** can include microprocessors, CPUs or the like, such as general or special purpose microprocessors operable to execute programming instructions or micro-control code associated with operation of the linear compressor **100**. Additionally, the controller **800** may also include a communications module **806** to facilitate communications between the controller and the various components of the refrigerator appliance **10**. An interface can include one or more circuits, terminals, pins, contacts, conductors, or other components for sending and receiving control signals. Moreover, the controller **800** may include a sensor interface **808** (e.g., one or more analog-to-digital converters) to permit signals transmitted from the temperature probe(s) **810** to be converted into signals that can be understood and processed by the processor(s) **802**. The controller **800** may furthermore optionally receive a second temperature signal(s) from the thermistor(s) **812** configured to generate one or more second temperature signals representative of the actual temperature of the item or the chamber.

Alternatively, the controller **800** may be constructed without using a microprocessor, e.g., using a combination of discrete analog and/or digital logic circuitry (such as switches, amplifiers, integrators, comparators, flip-flops, AND gates, and the like) to perform control functionality instead of relying upon software.

The inner back iron **142** further includes an outer cylinder **146** and an inner sleeve **148**. The outer cylinder **146** defines the outer surface of inner back iron **142** and also has an inner surface positioned opposite the outer surface of outer cylinder **146**. The inner sleeve **148** is positioned on or at inner surface of outer cylinder **146**. A first interference fit between outer cylinder **146** and inner sleeve **148** may couple or secure outer cylinder **146** and inner sleeve **148** together. In alternative exemplary embodiments, inner sleeve **148** may be welded, glued, fastened, or connected via any other suitable mechanism or method to outer cylinder **146**.

The outer cylinder **146** may be constructed of or with any suitable material. For example, outer cylinder **146** may be constructed of or with a plurality of (e.g., ferromagnetic) laminations. The laminations are distributed along the circumferential direction C in order to form outer cylinder **146** and are mounted to one another or secured together, e.g., with rings pressed onto ends of the laminations. The outer cylinder **146** may define a recess that extends inwardly from the outer surface of outer cylinder **146**, e.g., along the radial direction R. The driving magnet **144** is positioned in the recess on outer cylinder **146**, e.g., such that the driving magnet **144** is inset within outer cylinder **146**.

The linear compressor **100** also includes a plurality of planar springs **150**. Each planar spring **150** may be coupled to a respective end of inner back iron **142**, e.g., along the axial direction A. During operation of driving coil **124**, planar springs **150** support inner back iron **142**. In particular, the inner back iron **142** is suspended by planar springs **150** within the stator or the motor of the linear compressor **100** such that motion of inner back iron **142** along the radial direction R is hindered or limited while motion along the axial direction A is relatively unimpeded. Thus, the planar springs **150** may be substantially stiffer along the radial direction R than along the axial direction A. In such a manner, planar springs **150** can assist with maintaining a uniformity of the air gap between driving magnet **144** and driving coil **124**, e.g., along the radial direction R, during operation of the motor and movement of inner back iron **142** on the axial direction A. The planar springs **150** can also assist with hindering side pull forces of the motor from transmitting to piston **130** and being reacted in cylinder **117** as a friction loss.

A flex mount **160** is mounted to and extends through inner back iron **142**. In particular, the flex mount **160** is mounted to inner back iron **142** via inner sleeve **148**. Thus, the flex mount **160** may be coupled (e.g., threaded) to inner sleeve **148** at the middle portion of inner sleeve **148** and/or flex mount **160** in order to mount or fix flex mount **160** to inner sleeve **148**. The flex mount **160** may assist with forming a coupling **162**. The coupling **162** connects inner back iron **142** and piston **130** such that motion of inner back iron **142**, e.g., along the axial direction A, is transferred to piston **130**.

The coupling **162** may be a compliant coupling that is compliant or flexible along the radial direction R. In particular, coupling **162** may be sufficiently compliant along the radial direction R such that little or no motion of inner back iron **142** along the radial direction R is transferred to piston **130** by coupling **162**. In such a manner, side pull forces of the motor are decoupled from piston **130** and/or cylinder **117** and friction between piston **130** and cylinder **117** may be reduced.

As may be seen in the figures, the piston head **132** of piston **130** has a piston cylindrical side wall **170**. The cylindrical side wall **170** may extend along the axial direction A from piston head **132** towards inner back iron **142**. An outer surface of cylindrical side wall **170** may slide on cylinder **117** at chamber **118** and an inner surface of cylindrical side wall **170** may be positioned opposite the outer surface of cylindrical side wall **170**. Thus, the outer surface of cylindrical side wall **170** may face away from a center of cylindrical side wall **170** along the radial direction R, and the inner surface of cylindrical side wall **170** may face towards the center of cylindrical side wall **170** along the radial direction R.

The flex mount **160** extends between a first end portion **172** and a second end portion **174**, e.g., along the axial direction A. According to an exemplary embodiment, the inner surface of cylindrical side wall **170** defines a ball seat **176** proximate first end portion. In addition, coupling **162** also includes a ball nose **178**. Specifically, for example, the ball nose **178** is positioned at first end portion **172** of flex mount **160**, and ball nose **178** may contact flex mount **160** at first end portion **172** of flex mount **160**. In addition, ball nose **178** may contact piston **130** at ball seat **176** of piston **130**. In particular, ball nose **178** may rest on ball seat **176** of piston **130** such that ball nose **178** is slidable and/or rotatable on ball seat **176** of piston **130**. For example, ball nose **178** may have a frusto-spherical surface positioned against ball seat **176** of piston **130**, and ball seat **176** may be shaped complementary to the frusto-spherical surface of ball nose **178**. The frusto-spherical surface of ball nose **178** may slide and/or rotate on ball seat **176** of piston **130**.

Relative motion between the flex mount **160** and the piston **130** at the interface between ball nose **178** and ball seat **176** of piston **130** may provide reduced friction between piston **130** and cylinder **117**, e.g., compared to a fixed connection between flex mount **160** and piston **130**. For example, when an axis on which piston **130** slides within cylinder **117** is angled relative to the axis on which inner back iron **142** reciprocates, the frusto-spherical surface of ball nose **178** may slide on ball seat **176** of piston **130** to reduce friction between piston **130** and cylinder **117** relative to a rigid connection between inner back iron **142** and piston **130**.

Further, as shown, the flex mount **160** is connected to the inner back iron **142** away from first end portion **172** of flex mount **160**. For example, flex mount **160** may be connected to inner back iron **142** at second end portion **174** of flex mount **160** or between first and second end portions **172**, **174** of flex mount **160**. Conversely, the flex mount **160** is positioned at or within piston **130** at first end portion **172** of flex mount **160**, as discussed in greater detail below.

In addition, the flex mount **160** includes a tubular wall **190** between inner back iron **142** and piston **130**. A channel **192** within tubular wall **190** is configured for directing compressible fluid, such as refrigerant or air, though flex mount **160** towards piston head **132** and/or into piston **130**. Inner back iron **142** may be mounted to flex mount **160** such that inner back iron **142** extends around tubular wall **190**, e.g., at the middle portion of flex mount **160** between first and second end portions **172**, **174** of flex mount **160**. Channel **192** may extend between first and second end portions **172**, **174** of flex mount **160** within tubular wall **190** such that the compressible fluid is flowable from first end portion **172** of flex mount **160** to second end portion **174** of flex mount **160** through channel **192**. In such a manner, compressible fluid may flow through inner back iron **142** within flex mount **160** during operation of the linear compressor **100**. A muffler **194** may be positioned within channel **192** within tubular wall **190**, e.g., to reduce the noise of compressible fluid flowing through channel **192**.

The piston head **132** also defines at least one opening **196**. Opening **196** of piston head **132** extends, e.g., along the axial direction A, through piston head **132**. Thus, the flow of fluid may pass through piston head **132** via opening **196** of piston head **132** into chamber **118** during operation of the linear compressor **100**. In such a manner, the flow of fluid (that is compressed by piston head **132** within chamber **118**) may flow within channel **192** through flex mount **160** and inner back iron **142** to piston **130** during operation of the linear compressor **100**. As explained above, suction valve **128** (FIGS. **6-7**) may be positioned on piston head **132** to regulate the flow of compressible fluid through opening **196** into chamber **118**.

Referring still to FIGS. **3** through **7**, the linear compressor **100** may also include a lubrication system **200** for circulating a lubricant, e.g., such as oil, through the working or moving components of the linear compressor **100** to reduce friction, improve efficiency, etc. For example, as shown, the housing **102** may generally defines a sump **202** which is configured for collecting oil. Specifically, the sump **202** may be defined in the bottom portion of lower housing **104**. The lubrication system **200** further includes a pump **206** for continuously circulating oil through components of the linear compressor **100** which need lubrication.

As also illustrated in the figures, the linear compressor **100** may include a suction inlet **220** for receiving a flow of refrigerant. Specifically, as shown, the suction inlet **220** may be defined on the housing **102** (e.g., such as on lower housing **104**), and may be configured for receiving a refrigerant supply conduit to provide refrigerant to the cavity **108**. As explained above, the flex mount **160** includes tubular wall **190**, which defines channel **192** for directing compressible fluid, such as refrigerant gas, through flex mount **160** towards piston head **132**. In this manner, desirable flow path of refrigerant gas is through suction inlet **220**, through channel **192**, through opening **196**, and into chamber **118**. Suction valve **128** may block opening **196** during a compression stroke and a discharge valve **116** may permit the compressed gas to exit chamber **118** when the desired pressure is reached.

FIG. **9** depicts a flow diagram of an example method **900** for operating a linear compressor of an appliance according to example embodiments of the present disclosure. More particularly, the method **900** can be implemented to control a linear compressor of a refrigerator appliance (e.g., as depicted in FIGS. **1-2**) using a single-phase vector-like control scheme (e.g., as depicted in FIGS. **11**A-**11**B). FIG. **9** depicts steps performed in a particular order for purposes of illustration and discussion. Those of ordinary skill in the art, using the disclosures provided herein, will understand that various steps of any of the methods described herein can be omitted, expanded, performed simultaneously, rearranged, and/or modified in various ways without deviating from the scope of the present disclosure. Furthermore, various steps (not illustrated) can be performed without deviating from the scope of the present disclosure. Additionally, method **900** is generally discussed with reference to the refrigerator appliance **10** described above with reference to FIGS. **1-2** and the linear compressor **100** described above with reference to FIGS. **3-8**. However, it should be understood that aspects of the present method **900** can find application with any suitable appliance and/or linear compressor.

As noted above, the method **900** provides a method for operating a linear compressor (e.g., linear compressor **100**) of an appliance (e.g., refrigerator appliance **10**). The method **900** can include, at (**902**), operating a motor of the linear compressor in order to drive a rotor of the motor. In some embodiments, the linear compressor can include a single-phase motor such as, e.g., a single-phase linear motor. More particularly, a controller can be configured to implement a control scheme (e.g., control scheme **1000**) to operate a motor of the linear compressor (e.g., linear compressor **100**) in order to drive a rotor (e.g., rotor coupled to piston **130**) of the motor.

As an illustrative example, FIG. **10** depicts an example control scheme operable to control a linear compressor (e.g., linear compressor **100**) of an appliance (e.g., refrigerator appliance **10**) according to example embodiments of the present disclosure. A controller can be operably coupled to the motor of the linear compressor and can be configured to implement a control scheme **1000** to operate a motor of the linear compressor in order to drive a rotor (e.g., rotor coupled to piston **130**) of the motor. More particularly, the control scheme **1000** can include a field-weakening controller **1002**, a stroke controller **1004** (e.g., clearance controller **1004**), and a capacity controller **1006** (e.g., extension controller **1006**). As discussed in more detail below, the field-weakening controller **1002** can be configured to control a d-axis current component (I_d*) of the motor, the stroke controller **1004** (e.g., clearance controller **1004**) can be configured to control a q-axis current component (I_q*) of the motor, and the capacity controller **1006** (e.g., extension controller **1006**) can be configured to control a DC current component (I_0*) of the motor. Furthermore, as will be discussed in more detail below with reference to FIGS. **11**A-**11**B, the control scheme **1000** can further include a single-phase vector-like controller **1008**. The single-phase vector-like controller **1008** can be operatively coupled to the field-weakening controller **1002**, the stroke controller **1004**, and the capacity controller **1006**. The control scheme **1000** can further include an inverter **1010** configured to supply a variable frequency waveform to the motor. Furthermore, as will be discussed in greater detail below, the single-phase vector-like controller **1008** can be configured to determine the requisite voltage needed by the inverter **1010** in order to drive the motor based at least in part on the d-axis current component output by the field-weakening controller **1002**, the q-axis current component output by the stroke controller **1004**, and the DC current component output by the capacity controller **1006**.

As noted above, in some embodiments, the motor can be a single-phase linear motor. The single-phase linear motor can have a stator and a rotor. The rotor can be operatively coupled to a piston (e.g., a reciprocating piston) that compresses gas when operated. In some embodiments, the piston can be fitted with springs in order to allow resonant oscillation to facilitate the compression by the piston.

A total magnetic flux (λ) produced by the motor includes a rotor component (λ_r) and a stator component (λ_s). When the rotor is centered in the stator (i.e., x=0), total rotor flux (λ_r) linked to the stator is zero. When the rotor moves forward and/or backward from the center of the stator, the rotor flux (λ_r) increases and decreases, respectively, in a linear manner. Moreover, windings of the stator produce a stator flux proportional to a winding current in the stator winding (I) by the inductance (L). More particularly, the total flux (λ) is given by:

$$\lambda_r \approx \alpha x$$

$$\lambda_s = LI$$

$$\lambda = \lambda_s + \lambda_r = LI + \alpha x$$

When the total flux (λ) changes, an electromotive force voltage (EMF) is produced in the motor. More particularly, the EMF (¿) is given by:

$$\varepsilon = \dot{\lambda} = L\dot{I} + a\dot{x}$$

where $\dot{\lambda}$ is a time derivative of the total flux, $\dot{I}$ is a time derivative of the stator winding current, and $\dot{x}$ is a time derivative of the rotor position. Furthermore, ax represents a back-EMF voltage (back-EMF) of the motor.

Taking into account the above-mentioned equation for the total EMF, a total voltage (V) for the motor is given by:

$$V = RI + L\dot{I} + a\dot{x}$$

As noted above, in some embodiments, the piston can be fitted with springs to allow the resonant oscillation to facilitate the compression by the piston. More particularly, the piston oscillates in an approximately sinusoidal manner. The sinusoidal oscillation of the piston is given by:

$$x(t) = x_1 \sin\theta + x_0$$

where θ represents a phase of the sine wave, x_1 represents an amplitude of the sine wave, and x_0 represents an offset of a midpoint of oscillation with respect to the center of the stator winding. Those of ordinary skill in the art will understand that the asymmetric force of gas compression can induce a positive offset in the position of the sine waveform.

Taking the above-equation, the time derivative of the rotor position (for use in the total motor voltage equation) can be derived. More particularly, the time derivative of the rotor position is given by:

$$\dot{x}(t) = \omega x_1 \cos\theta$$

where $\omega = \dot{\theta}$. As used herein, "velocity equation" refers to the above-described equation.

Referring still the FIG. **10**, the control scheme **1000** can be configured to control the amplitude of the motor current (I) and the relative phase between the motor current (I) and the back-EMF. Those of ordinary skill in the art will understand that current which is in-phase with the back-EMF generates a net force which increases stroke and capacity of the linear compressor. Likewise, those of ordinary skill in the art will understand that current which is out-of-phase with the back-EMF does not generate a net force but does affect the total voltage (V) needed by the inverter (e.g., inverter **1010**) to drive the motor. In this way, the out-of-phase current can be useful for field-weakening operations.

The phase of the rotor flux (θ) can be used to define an in-phase component and an out-of-phase component of the sinusoidal motor current (I). The total motor current (I) and its time derivative ($\dot{I}$) are given by:

$$I(t) = I_d \sin\theta + I_q \cos\theta$$

$$\dot{I}(t) = \omega I_d \cos\theta - \omega I_q \sin\theta$$

where I_d is the amplitude of the current component that is out-of-phase with the back-EMF (i.e., in-phase with the rotor flux) and Ig is the amplitude of the current component that is in-phase with the back-EMF (i.e., out-of-phase with the rotor flux).

Thus, substituting I(t), $\dot{I}$(t), and $\dot{x}$(t) into the total voltage (V) for the motor can be represented:

$$V = RI_d \sin\theta + RI_q \cos\theta + L\omega I_d \cos\theta - \omega I_q \sin\theta + \alpha\omega x_1 \cos\theta$$

$$V = V_d \sin\theta + V_q \cos\theta$$

where $V_d = RI_d - \omega LI_q$ and $V_q = RI_q + \omega LI_d + \omega\alpha x_1$. Those of ordinary skill in the art will understand that the above-described equations directly parallel the dq voltage equations for a three-phase permanent magnet synchronous motor (PMSM). In this way, the present disclosure provides a single-phase vector-like control scheme that is analogous to a field-oriented control scheme commonly used in three-phase motors such as, e.g., PMSMs and brushless DC (BLDC) motors.

Referring still to FIG. **10**, the control scheme **1000** can include a field-weakening controller **1002** configured to control the d-axis current component of the motor current, a stroke controller **1004** (e.g., clearance controller **1004**) configured to control the q-axis current component of the motor current, and a capacity controller **1006** (e.g., extension controller **1006**) configured to control the DC current component of the motor current. More particularly, the field-weakening controller **1002** can be configured to determine a target d-axis current component (I_d*), the stroke controller **1004** can be configured to determine a target q-axis current component (I_q*), and the capacity controller **1006** can be configured to control a target DC current component (I_0*).

The target d-axis current component (I_d*), target q-axis current component (I_q*), and target DC current component (I_0*) can then be passed to the single-phase vector-like controller **1008**. As noted above, the single-phase vector-like controller **1008** can be configured to determine the requisite voltage (V) required by the inverter **1010** in order to drive the motor. Furthermore, the single-phase vector-like controller **1008** can determine the total voltage needed by the inverter **1010** based at least in part on the target d-axis current component (I_d*), the target q-axis current component (I_q*), and the target DC current component (I_0*).

Returning to FIG. **9**, the method **900** can include, at (**904**), obtaining one or more feedback measurements of one or more electrical characteristics of the motor. More particularly, the controller can be configured to obtain one or more feedback measurements (e.g., current feedback, voltage feedback, etc.) of one or more electrical characteristics (e.g., current, voltage, magnetic flux, etc.) of the motor.

As an illustrative example, referring to FIG. **10**, the controller can be configured to obtain one or more feedback measurements (e.g., current feedback, voltage feedback, etc.) of one or more electrical characteristics (e.g., current, voltage, magnetic flux, etc.) of the motor. In some embodiments, the control scheme **1000** can include a sensored feedback system (not shown) configured to obtain the one or more feedback measurements. For instance, in some embodiments, the control scheme **1000** can include a sensor (e.g., encoder, Hall effect sensor, etc.) configured to obtain a velocity signal and a phase-locked loop (PLL) configured to extract the frequency and phase from the velocity signal.

Referring still to FIG. **10**, in other embodiments, the control scheme **1000** can include a sensorless feedback system (not shown) configured to obtain the one or more feedback measurements. As used herein, a "sensorless" feedback system can refer to any feedback system operable to determine data indicative of a position or a speed of the motor without a position sensor or a speed sensor. For instance, in some embodiments, the control scheme **1000** can include, e.g., an observer (not shown) configured to estimate back-EMF and/or velocity based at least in part on voltage and/or current signals of the motor. The back-EMF signal and/or the velocity signal can then be fed into a PLL (as described above) to extract the frequency and phase of the corresponding signal.

Referring to FIG. **9**, the method **900** can include, at (**906**), controlling the motor of the linear compressor using a single-phase vector-like control scheme based at least in part on the one or more feedback measurements obtained at (**904**). More particularly, the controller can be configured to implement a single-phase vector-like control scheme (e.g., control scheme **1000**) to control the motor of the linear compressor (e.g., linear compressor **100**) based at least in part on the one or more feedback measurements obtained by the controller.

As an illustrative example, referring to FIG. **10**, the single-phase vector-like controller **1008** can be configured to implement a single-phase vector-like control scheme in order to control the motor of the linear compressor (e.g., linear compressor **100**). In some embodiments, the controller can be configured to implement single-phase tracking control. For instance, FIG. **11**A depicts an example embodiment of the single-phase vector-like controller **1008** implementing single-phase tracking control. As will be discussed in greater detail below with reference to FIG. **13**, the single-phase vector-like controller **1008** can be configured to determine a target current trajectory (I*).

The target current trajectory can be based on the target d-axis current component (e.g., d-axis current setpoint (I_d*)) from the field-weakening controller **1002**, the target q-axis current component (e.g., q-axis current setpoint (I_d)) from the stroke controller **1004**, and the target DC current component (e.g., DC current setpoint (I_q*)) from the capacity controller **1006**. The target current trajectory can also be based on a feedback measurement of a phase angle of rotor (e.g., rotor coupled to piston **130**) magnetic flux obtained at (**904**). The control scheme **1000** can be configured to determine an instantaneous motor voltage (V) necessary to force the actual motor current (I(t)) to track the target current trajectory (I*) in real time.

Similarly, in other embodiments, the controller can be configured to implement single-phase DQ**0** control. For instance, FIG. **11**B depicts an example embodiment of the single-phase vector-like controller **1008** described above with reference to FIG. **10** implementing single-phase DQ**0** control. As will be discussed in greater detail below with reference to FIGS. **14-15**, the single-phase vector-like controller **1008** can be configured to determine an actual d-axis current component (I_d), an actual q-axis current component (I_q), and an actual DC current component (I_0) based at least in part on the current feedback measurements (I) obtained at (**904**). The control scheme **1000** can then use the actual d-axis current component (I_d), the actual q-axis current component (I_q), and the actual DC current component (I_0) to determine an instantaneous voltage (V) for the motor.

FIG. **12** depicts an example method for controlling the motor of the linear compressor (e.g., linear compressor **100**) using a single-phase vector-like control scheme (e.g., implemented by single-phase vector-like controller **1008** of control scheme **1000**) based at least in part on the one or more feedback measurements obtained at (**904**). Furthermore, FIG. **12** depicts steps performed in a particular order for purposes of illustration and discussion. Those of ordinary skill in the art, using the disclosures provided herein, will understand that various steps of any of the methods described herein can be omitted, expanded, performed simultaneously, rearranged, and/or modified in various ways without deviating from the scope of the present disclosure. Furthermore, various steps (not illustrated) can be performed without deviating from the scope of the present disclosure.

Referring to FIG. **12** at (**1202**), the controller (e.g., via control scheme **1000**) can be configured to adjust a d-axis current setpoint based at least in part on the one or more feedback measurements obtained at (**904**). As used herein, "d-axis current setpoint" refers to the target d-axis current component (I_d*) discussed above with reference to FIG. **10**. For instance, as noted above with reference to FIG. **10**, the field-weakening controller **1002** can be configured to control the target d-axis current component of the motor. More particularly, in some embodiments, the field-weakening controller **1002** can be configured to adjust the setpoint of the d-axis current component (I_d*) based at least in part on the one or more feedback measurements (e.g., current feedback) obtained at (**904**). The adjusted d-axis current setpoint (I_d*) can then be used by the single-phase vector-like controller **1008**, in part, to determine the total voltage required by the inverter **1010**. Those of ordinary skill in the art will understand that the d-axis current setpoint can be adjusted in any suitable manner without deviating from the scope of the present disclosure.

Referring to FIG. **12** at (**1204**), the controller (e.g., via control scheme **1000**) can be configured to adjust a q-axis current setpoint based at least in part on the one or more feedback measurements obtained at (**904**). As used herein, "q-axis current setpoint" refers to the target q-axis current component (I) discussed above with reference to FIG. **10**. For instance, as noted above with reference to FIG. **10**, the stroke controller **1004** can be configured to control the q-axis current component of the motor. More particularly, in some embodiments, the stroke controller **1004** can be configured to adjust the setpoint of the q-axis current component (I_q*) based at least in part on the one or more feedback measurements (e.g., current feedback) obtained at (**904**). The adjusted q-axis current setpoint (I_q*) can then be used by the single-phase vector-like controller **1008**, in part, to determine the total voltage required by the inverter **1010**. Those of ordinary skill in the art will understand that the q-axis current setpoint can be adjusted in any suitable manner without deviating from the scope of the present disclosure.

Referring to FIG. **12** at (**1206**), the controller (e.g., via control scheme **1000**) can be configured to adjust a DC current setpoint based at least in part on the one or more feedback measurements obtained at (**904**). As used herein, "DC current setpoint" refers to the target DC current component (I_0*) discussed above with reference to FIG. **10**. For instance, as noted above with reference to FIG. **10**, the capacity controller **1006** can be configured to control the DC current component of the motor. More particularly, in some embodiments, the capacity controller **1006** can be configured to adjust the setpoint of the DC current component (I_0*) based at least in part on the one or more feedback measurements (e.g., current feedback) obtained at (**904**). The adjusted DC current setpoint (I_0*) can then be used by the single-phase vector-like controller **1008**, in part, to determine the total voltage required by the inverter **1010**. Those of ordinary skill in the art will understand that the DC current setpoint can be adjusted in any suitable manner without deviating from the scope of the present disclosure.

FIG. **13** depicts an example method for controlling the motor of the linear compressor (e.g., linear compressor **100**) using a single-phase vector-like control scheme (e.g., implemented by single-phase vector-like controller **1008** of control scheme **1000**) based at least in part on the one or more feedback measurements obtained at (**904**). Furthermore, FIG. **13** depicts steps performed in a particular order for purposes of illustration and discussion. Those of ordinary skill in the art, using the disclosures provided herein, will understand that various steps of any of the methods described herein can be omitted, expanded, performed simultaneously, rearranged, and/or modified in various ways without deviating from the scope of the present disclosure. Furthermore, various steps (not illustrated) can be performed without deviating from the scope of the present disclosure.

Referring to FIG. **13** at (**1302**), the controller (e.g., via control scheme **1000**) can be configured to determine a target current trajectory. For instance, referring to FIG. **11**A, the control scheme **1000** can be configured to combine the d-axis current setpoint determined at (**1202**) with the phase of the rotor flux (θ) at **1102**. The control scheme **1000** can be further configured to combine the q-axis current setpoint determined at (**1204**) with the phase of the rotor flux (θ) at **1104**. The control scheme **1000** can be further configured to sum, at **1106**, the current values determined at **1102** and **1104** with the DC current setpoint determined at (**1206**). Thus, the control scheme **1000** can be configured to determine the target current trajectory at **1106**. The target current trajectory (I*) is given by:

$$I^* = I_d^*\sin\theta + I_q^*\cos\theta + I_0^*$$

Referring to FIG. **13** at (**1304**), the controller (e.g., via control scheme **1000**) can be configured to determine a trajectory difference (e.g., a trajectory error) between the target current trajectory determined at (**1302**) and the one or more feedback measurements obtained at (**904**). For instance, referring to FIG. **11**A, the control scheme **1000** can be configured to determine a trajectory difference at **1108**. More particularly, the control scheme **1000** can be configured to determine a difference between the target current trajectory (I*) determined at **1106** and the current feedback measurements (I). As noted above with reference to FIG. **10**, the current feedback measurements (I) can be obtained by a sensored feedback system or a sensorless feedback system without deviating from the scope of the present disclosure.

Referring to FIG. **13** at (**1306**), responsive to determining the trajectory difference at (**1304**), the controller (e.g., via control scheme **1000**) can be configured to adjust a voltage setpoint based at least in part on the trajectory difference (e.g., trajectory error) determined at (**1304**). For instance, referring to FIG. **11**A, the control scheme **1000** can include a current controller **1110** configured to adjust the total voltage setpoint (V) based at least in part on the trajectory difference determined at **1108**. The total voltage setpoint (V) can then be sent to the inverter **1010**.

FIG. **14** depicts an example method for controlling the motor of the linear compressor (e.g., linear compressor **100**) using a single-phase vector-like control scheme (e.g., implemented by single-phase vector-like controller **1008** of control scheme **1000**) based at least in part on the one or more feedback measurements obtained at (**904**). Furthermore, FIG. **14** depicts steps performed in a particular order for purposes of illustration and discussion. Those of ordinary skill in the art, using the disclosures provided herein, will understand that various steps of any of the methods described herein can be omitted, expanded, performed simultaneously, rearranged, and/or modified in various ways without deviating from the scope of the present disclosure. Furthermore, various steps (not illustrated) can be performed without deviating from the scope of the present disclosure.

More particularly, as noted above with reference to FIG. **11**B, the control scheme **1000** can be configured to obtain current feedback measurements (I) and rotor flux feedback measurements. Referring to FIG. **11**B, the control scheme **1000** can include a second-order generalized integrator **1150** (SOGI **1150**) and a Park transform **1152** which, when used in tandem, can be configured to determine the actual d-axis current component (I_d), the actual q-axis current component (I_q), and the actual DC current component (I_0) of the current feedback measurements (I). In order to determine the actual d-axis current component (I_d), the actual q-axis current component (I_q), and the actual DC current component (I_0) of the current feedback measurements, the SOGI **1150** uses the frequency component (w) of the rotor flux feedback measurements, and the Park Transform **1152** uses the phase component (**0**) of the rotor flux feedback measurements.

Referring to FIG. **14** at (**1402**), the controller (e.g., via control scheme **1000**) can be configured to determine a field-weakening current difference (e.g., a field-weakening current error) between the d-axis current setpoint determined at (**1202**) and the actual d-axis current component of the current feedback measurements obtained at (**904**). For instance, referring to FIG. **11**B, the control scheme **1000** can be configured to determine a field-weakening difference **1154** between the d-axis current setpoint (I_d*) and the actual d-axis current component (I_d) of the current feedback measurements (I).

Referring to FIG. **14** at (**1404**), the controller (e.g., via control scheme **1000**) can be configured to determine a stroke current difference (e.g., a stroke current error) between the q-axis current setpoint determined at (**1204**) and the actual q-axis current component of the current feedback measurements obtained at (**904**). For instance, referring to FIG. **11**B, the control scheme **1000** can be configured to determine a stroke difference **1156** between the q-axis current setpoint (I_q*) and the actual q-axis current component (I_q) of the current feedback measurements (I).

Referring to FIG. **14** at (**1406**), the controller (e.g., via control scheme **1000**) can be configured to determine a capacity current difference (e.g., a capacity current error) between the DC current setpoint determined at (**1206**) and the actual DC current component of the current feedback measurements obtained at (**904**). For instance, referring to FIG. **11**B, the control scheme **1000** can be configured to determine a capacity difference **1158** between the DC current setpoint (I_0*) and the actual DC current component (I_0) of the current feedback measurements (I).

FIG. **15** depicts an example method for controlling the motor of the linear compressor (e.g., linear compressor **100**) using a single-phase vector-like control scheme (e.g., implemented by single-phase vector-like controller **1008** of control scheme **1000**) based at least in part on the one or more feedback measurements obtained at (**904**). Furthermore, FIG. **15** depicts steps performed in a particular order for purposes of illustration and discussion. Those of ordinary skill in the art, using the disclosures provided herein, will understand that various steps of any of the methods described herein can be omitted, expanded, performed simultaneously, rearranged, and/or modified in various ways without deviating from the scope of the present disclosure. Furthermore, various steps (not illustrated) can be performed without deviating from the scope of the present disclosure.

Referring to FIG. **15** at (**1502**), the controller (e.g., via control scheme **1000**) can be configured to adjust a d-axis voltage setpoint based at least in part on the field-weakening current difference (e.g., field-weakening current error) determined at (**1402**). For instance, referring to FIG. **11**B, the control scheme **1000** can include a field-weakening current proportional-integral controller (PI) **1160** (e.g., la Current PI **1160**). The field-weakening current PI **1160** can be configured to determine the d-axis voltage component (V_d) based at least in part on the field-weakening current difference **1154** determined at (**1402**). As used herein, "d-axis voltage setpoint" can refer to the d-axis voltage component (V_d). Those of ordinary skill in the art will understand that the d-axis voltage setpoint can be adjusted in any suitable manner without deviating from the scope of the present disclosure.

Referring to FIG. **15** at (**1504**), the controller (e.g., via control scheme **1000**) can be configured to adjust a q-axis voltage setpoint based at least in part on the stroke current difference (e.g., stroke current error) determined at (**1404**). For instance, referring to FIG. **11**B, the control scheme **1000** can include a stroke current PI **1162** (e.g., I_q Current PI **1162**). The stroke current PI **1162** can be configured to determine the q-axis voltage component (V_q) based at least in part on the stroke current difference **1156** determined at (**1404**). As used herein, "q-axis voltage setpoint" can refer to the q-axis voltage component (V_q). Those of ordinary skill in the art will understand that the q-axis voltage setpoint can be adjusted in any suitable manner without deviating from the scope of the present disclosure.

Referring to FIG. **15** at (**1506**), the controller (e.g., via control scheme **1000**) can be configured to adjust a DC voltage setpoint based at least in part on the capacity current difference (e.g., capacity current error) determined at (**1406**). For instance, referring to FIG. **11**B, the control scheme **1000** can include a capacity current PI **1164** (e.g., I_0 Current PI **1164**). The capacity current PI **1164** can be configured to determine the DC voltage component (V_0) based at least in part on the capacity current difference **1158** determined at (**1406**). As used herein, "DC voltage setpoint" can refer to the DC voltage component (V_0). Those of ordinary skill in the art will understand that the DC voltage setpoint can be adjusted in any suitable manner without deviating from the scope of the present disclosure.

Furthermore, referring to FIG. **11**B, the d-axis voltage setpoint determined at (**1502**) can be combined with the phase of the rotor flux (θ) at **1166**, and the q-axis voltage setpoint determined at (**1504**) can be combined with the phase of the rotor flux (θ) at **1168**. The voltage values determined at **1166** and **1168** can then be summed, at **1170**, with the DC voltage setpoint determined at (**1506**) to calculate the total voltage (V) necessary to drive the motor. The total voltage (V) can then be sent to the inverter **1010**. Accordingly, the total voltage determined at **1170** is given by:

$$V = V_d \sin\theta + V_q \cos\theta + V_0$$

While the present subject matter has been described in detail with respect to specific example embodiments thereof, it will be appreciated that those skilled in the art, upon attaining an understanding of the foregoing can readily produce alterations to, variations of, and equivalents to such embodiments. Accordingly, the scope of the present disclosure is by way of example rather than by way of limitation, and the subject disclosure does not preclude inclusion of such modifications, variations and/or additions to the present subject matter as would be readily apparent to one of ordinary skill in the art.

What is claimed is:

1. A method for operating a linear compressor of an appliance, the method comprising:

operating a motor of the linear compressor in order to drive a rotor of the motor,

obtaining, via a controller of the linear compressor, one or more feedback measurements of one or more electrical characteristics of the motor; and

controlling, based at least in part on the one or more feedback measurements, the motor of the linear compressor using a single-phase control scheme by:

adjusting, via the controller, a d-axis current setpoint based at least in part on the one or more feedback measurements;

adjusting, via the controller, a q-axis current setpoint based at least in part on the one or more feedback measurements; and

adjusting, via the controller, a DC current setpoint based at least in part on the one or more feedback measurements.

2. The method of claim **1**, wherein controlling the motor using the single-phase control scheme further comprises:

determining, via the controller, a target current trajectory;

determining, via the controller, a trajectory difference between the target current trajectory and the one or more feedback measurements; and

responsive to determining the trajectory difference, adjusting, via the controller, a voltage setpoint based at least in part on the trajectory difference.

3. The method of claim **1**, wherein controlling the motor using the single-phase control scheme further comprises:

determining, via the controller, a field-weakening current difference between the d-axis current setpoint and a d-axis current component of the feedback measurements;

determining, via the controller, a stroke current difference between the q-axis current setpoint and a q-axis current component of the feedback measurements; and

determining, via the controller, a capacity current difference between the DC current setpoint and a DC current component of the feedback measurements.

4. The method of claim 3, wherein controlling the motor using the single-phase control scheme further comprises:

adjusting, via the controller, a d-axis voltage setpoint based at least in part on the field-weakening current difference;

adjusting, via the controller, a q-axis voltage setpoint based at least in part on the stroke current difference; and

adjusting, via the controller, a DC voltage setpoint based at least in part on the capacity current difference.

5. The method of claim 1, wherein the one or more feedback measurements comprises measurements indicative of a phase angle of magnetic flux of the rotor.

6. The method of claim 5, wherein the controller comprises a sensored feedback system, the sensored feedback system configured to obtain one or more feedback measurements of the one or more electrical characteristics of the motor.

7. The method of claim 5, wherein the controller comprises a sensorless feedback system, the sensorless feedback system configured to obtain one or more feedback measurements of the one or more electrical characteristics of the motor.

8. The method of claim 7, wherein the sensorless feedback system comprises a back-electromotive force (back-EMF) observer.

9. A linear compressor defining an axial direction and a vertical direction, the linear compressor for an appliance comprising:

a cylindrical casing defining a compressor chamber;

a piston positioned within the compressor chamber and being movable along the axial direction;

a motor operably coupled to the piston; and

a controller operably coupled to the motor, the controller configured to:

operate the motor in order to drive a rotor of the motor;

obtain one or more feedback measurements of one or more electrical characteristics of the motor, the one or more feedback measurements comprising measurement indicative of a phase angle of magnetic flux of the rotor; and

control the motor using a single-phase control scheme based at least in part on the one or more feedback measurements.

10. The linear compressor of claim 9, wherein the controller is further configured to:

determine a target current trajectory;

determine a trajectory difference between the target current trajectory and the one or more feedback measurements; and

adjust a voltage setpoint based at least in part on the trajectory difference.

11. The linear compressor of claim 9, wherein the controller is further configured to:

adjust a d-axis current setpoint based at least in part on the one or more feedback measurements;

adjust a q-axis current setpoint based at least in part on the one or more feedback measurements; and

adjust a DC current setpoint based at least in part on the one or more feedback measurements.

12. The linear compressor of claim 11, wherein the controller is further configured to:

determine a field-weakening current difference between the d-axis current setpoint and a d-axis current component of the feedback measurements;

determine a stroke current difference between the q-axis current setpoint and a q-axis current component of the feedback measurements; and

determine a capacity current difference between the DC current setpoint and a DC current component of the feedback measurements.

13. The linear compressor of claim 12, wherein the controller is further configured to:

adjust a d-axis voltage setpoint based at least in part on the field-weakening current difference;

adjust a q-axis voltage setpoint based at least in part on the stroke current difference; and

adjust a DC voltage setpoint based at least in part on the capacity current difference.

14. The linear compressor of claim 9, wherein the piston is a reciprocating piston.

15. The linear compressor of claim 9, wherein the motor is a single-phase linear motor.

16. The linear compressor of claim 9, wherein the controller comprises a sensorless feedback system, the sensorless feedback system being configured to obtain the one or more feedback measurements of one or more electrical characteristics of the motor.

17. An appliance, comprising:

a cabinet defining an internal chamber;

a door rotatably mounted to the cabinet to provide selective access to the internal chamber;

a linear compressor, the linear compressor having a piston movable in a negative axial direction toward a compressor chamber and a positive axial direction away from the compressor chamber;

a motor operably coupled to the piston;

an inverter configured to supply a variable frequency waveform to the motor; and

a controller operably coupled to the motor, the controller configured to:

operate the motor in order to drive a rotor of the motor;

obtain one or more feedback measurements of one or more electrical characteristics of the motor;

determine a target current trajectory;

determine a trajectory difference between the target current trajectory and the one or more feedback measurements; and

adjust a voltage setpoint based at least in part on the one or more feedback measurements.

18. The appliance of claim 17, wherein:

the piston is a reciprocating piston; and

the motor is a single-phase linear motor.

19. The appliance of claim 17, wherein the appliance is a refrigerator appliance.

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